Synthesis S. Basak et al. Special Topic

Transition-Metal-Catalyzed C–H Arylation Using Organoboron Reagents

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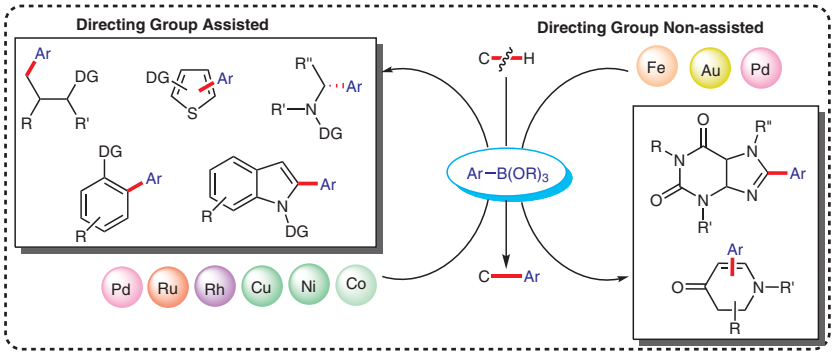

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Abstract Aryl rings are ubiquitous in the core of numerous natural product and industrially important molecules and thus their facile synthesis is of major interest in the scientific community and industry. Although multiple strategies enable access to these skeletons, metal-catalyzed C–H activation is promising due to its remarkable efficiency. Commercially available organoboron reagents, a prominent arylating partner in the cross-coupling domain, have also been utilized for direct arylation. Organoborons are bench-stable, inexpensive, and readily available coupling partners that promise regioselectivity, chemodivergence, cost-efficiency, and atom-economy without requiring harsh and forcing conditions. This critical, short review presents a summary of all major studies of arylation using organoborons in transition-metal catalysis since 2005.

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Key words transition-metal catalysis, C–H activation, arylation, organoboron, C–H functionalization, heterocycle

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1 Introduction

During the last century, synthetic chemistry has witnessed remarkable progress with the help of extraordinary synthetic toolkits. In this regard, synthetic strategies are desired to be regioselective, chemodivergent, atom-economic,

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and commercially viable. Myriad bioactive molecular architectures with aromatic rings are fundamental to naturally abundant molecules,[1,2] pharmacophores,[3] agrochemicals, and various other industrially relevant materials.[4] In the second half of the 20th century, cross-coupling reactions showed great promise for the almost frictionless creation of many semi-complex frameworks. Typical cross-coupling reactions require functionalized reactants and a stoichiometric amount of organometallic reagents for joining relatively simple structures. However, this phenomenal method fell short in the context of industrial realization. Cross-coupling reactions invariably require pre-functionalization of the substrate, which is highly undesirable especially for complex reactants. Accessing such starting reagents in many instances becomes quite difficult and demands multistep synthesis. Thus, commercial state-of-art viability demands straightforward protocols that are ecofriendly, cost-effective, step- and atom-economic. C–H activation is that emerging toolbox which can meet the versatile demands to displace cross-coupling.[5–7] The methods primarily deploy a catalytic amount of metals for oxidative coupling between C–H bonds and other reactant partners to construct C–C and C–N bonds with ease. Direct functionalization can obviate most of the shortcomings of conventional cross-coupling reactions with a comparable degree of efficiency and has high potential for providing access to newer complex molecules. However, the presence of multiple C–H bonds in most feedstocks, attainment of the desired degree of regioselectivity, and avoidance of overfunctionalization become challenging.[8] Towards this, directing groups have offered improvements although their introduction and elimination add extra steps. Partial circumvention of these drawbacks requires direct functionalization strategies to take inspiration from cross-coupling methods without sacrificing environmental benignity and, therefore, organoboron reagents[9] transcended from the realm of conventional cross-coupling to the expanding domain of C–H functionalization. Organoborons are bench-stable, inexpensive, and readily available coupling partners that promise regioselectivity without requiring harsh and forcing conditions. This class of reagents is widely used in Suzuki–Miyaura coupling[10] and since 2000, a growing number of direct C–H functionalization protocols have exploited their reactivity and availability.[11–13] A number of different metals has been used as catalysts and these demonstrated exceptional efficacy in producing high yields, even in on a gram scale. This critical and comprehensive short review covers all major studies that have contributed to the extension of the functionalization toolbox deploying both noble as well as earth-abundant metals in C–H arylation reactions utilizing organoboron reagents since 2005. On a specific note, arylation without the assistance of a directing group is discussed first to highlight the future need for step-economic strategies.

2 Arylation without Directing Group Assistance

The application of arylborons has shown impact in C–H coupling reactions that require little influence of external directing groups (DGs). Such transformations required exquisite ligands and catalysts to produce yields on a par with directing group assisted methods, often leading to a trade-off between the cost of the reaction and its advantageous step-economy. Notably, it is difficult to classify methods as 'directing group non-assisted' since the presence of a heteroatom often leads to chelation at the molecular level but tracing their exact role on a macroscale is challenging. The methods discussed in this section have been categorized based on reported evidence and proposed mechanisms. Noble metals, like Pd, remain over-utilized in this segment although recently a few coin and earth-abundant metals have been used, showing promise for expansion with more inexpensive catalysts.

2.1 Palladium Catalysis

In 2008, the Georg group reported the arylation of (bi)cyclic enaminones at C-5 to produce 3-arylpiperidine motifs, skeletal components of pharmaceutically important molecules (Scheme 1).[8] They took advantage of the intrinsic nucleophilicity of the enaminone for the direct arylation, catalyzed by $Pd(OAc)_2$ at relatively mild 60 °C temperature. The utilization of lower amounts of pre-catalyst required longer time frames and the use of higher temperatures. They postulated that arylboronic acids could not be used due to their scant dissolution in the acidic co-solvent and instead potassium aryltrifluoroborates ($ArBF_3K$) were used. The slow addition of fluoroborate resulted in improved yields, which was attributed the suppression of homocoupling. Electron-rich and -poor fluoroborates coupled with monocyclic and bicyclic enaminones, and the former gave better yields compared to the later. The N-substituent on the enaminone had a role in electronically directing the reaction; *N*-alkyl enaminones (R^1 = Me, Bn) participated in reaction whereas *N*-Boc-protected enaminones remained unreactive.

The proposed mechanism (Scheme 1) is bimodal. Initially, in path A, **C1** electrophilically attacks C-5 of the enaminone to produce **C2**, which undergoes transmetalation to form **C3** and subsequent reductive elimination to generate the isolated product. Alternatively, in path B transmetalation followed by C–H activation leads to the common intermediate **C3**. There is as yet no conclusive proof to support these proposed routes.

Also in 2008, the Shi group reported the direct coupling between electron-rich heteroarenes with arylboronic acids under an oxygen atmosphere (Scheme 2).[14] Surprisingly, they observed that on variation of the terminal oxidant, the catalyst [$Pd(OAc)_2$] loading could be reduced without sacri-

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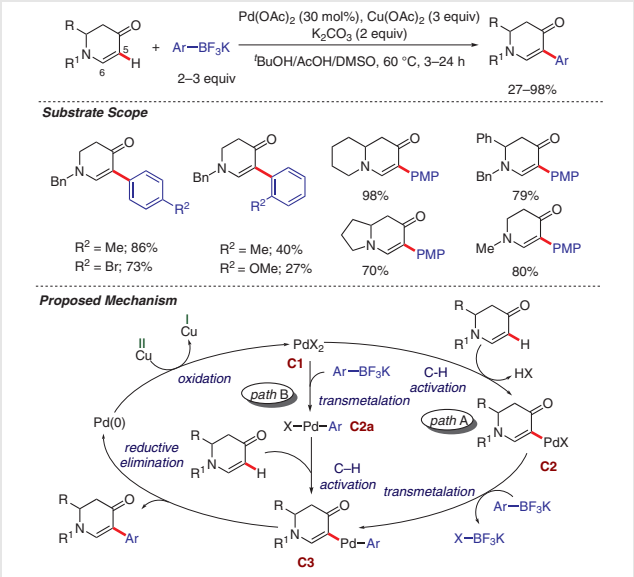
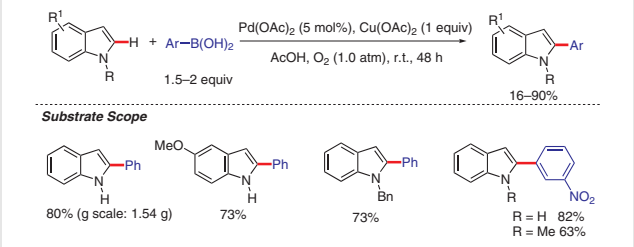


Scheme 1 Pd-catalyzed C–H arylation of (bi)cyclic enaminones at C-5 and proposed mechanism

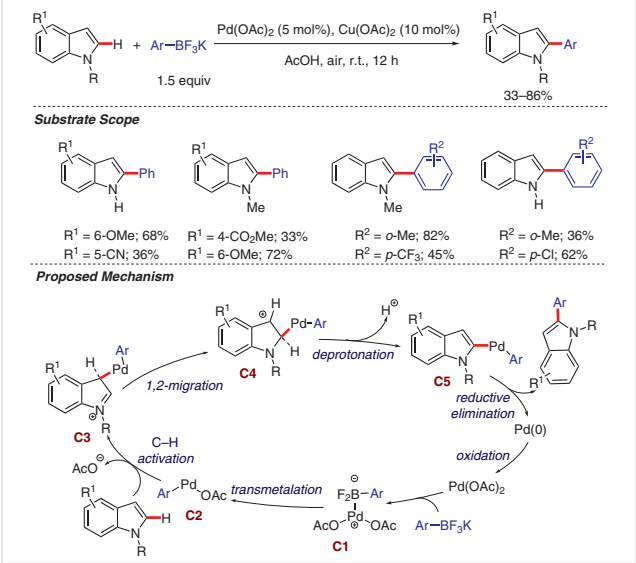
ficing the yields. *N*-(Un)substituted indoles, *N*-unsubstituted pyrroles, benzothiophene, and benzofuran were easily monoarylated in up to 87% yield. *N*-Alkylindoles gave productive outcomes whereas *N*-acetylindoles proved poor substrates for the protocol. Intriguingly, *N*-unsubstituted indole and 2-tolylboronic acid gave the corresponding product in 50% yield, and it was argued that here the low yield was due to steric hindrance. But in case of the coupling between *N*-methylindole and 2-tolylboronic acid, that gave 87% yield of the corresponding product, no role of steric hindrance was surmised. The reaction of unsubstituted 1*H*-indole with phenylboronic acid resulted in arylation in on a gram scale in 80% yield; recovery of both catalyst and acetic acid at the end was possible.

Zhang and co-workers investigated the Pd-catalyzed arylation of indoles at C-2 using potassium aryltrifluoroborates at room temperature under $Cu(OAc)_2$ oxidation (Scheme 3);[15] their conditions were similar to those used by the Shi group.[14] Electron-rich fluoroborates furnished better results compared to electron-deficient ones and a



Scheme 2 2-Arylindole synthesis through Pd-catalyzed C–H arylation of *N*-(un)substituted indoles

wide range of functionalities, such as F, Cl, CF_3, OMe, and Me, were tolerated. *N*-Substituted indoles also experienced little steric hindrance compared to *N*-unsubstituted indoles in this approach when potassium 2-tolyltrifluoroborate was utilized. Electronically impoverished indoles gave inferior results, whereas electron-rich ones engendered good yields. Notably, the presence of molecular oxygen was necessary as the use of 1 equivalent $Cu(OAc)_2$ alone was insufficient for the conversion and it remained largely inefficacious.



Scheme 3 2-Arylindole synthesis through Pd-Cu co-catalyzed C–H arylation of *N*-(un)substituted indoles and proposed mechanism

The proposed mechanism (Scheme 3) involves initial transmetalation; **C2** undergoes palladation slowly in the acidic media to form **C3** and its sluggish rate allows 1,2-migration of the catalyst from C-3 to C-2 of indole to generate **C4**. Formal deprotonation leads to **C5** and succeeding reductive elimination gives the product, followed by oxidation of Pd(0) to Pd(II) to complete the catalytic cycle.

In 2009, Su and co-workers reported the arylation of electron-poor polyfluoroarenes, which is difficult because of their chemical inertness (Scheme 4).[16] They added both a weak acid and base externally, rationalizing their need as homocoupling suppressant and C–H cleavage promotor, respectively. Electron-rich arylboronic acids containing NMe_2, Me, and OMe groups performed better compared to electron-poor ones, and polyfluorobenzenes containing Br, CN, Ac, and OMe functionalities were tolerated under the applied conditions. 2-Tolylboronic acid could not be coupled owing to steric factors experienced by the 2,6-fluoro groups. Mechanistic investigations revealed C–H cleavage probably occurs as the rate-determining step of the reaction.

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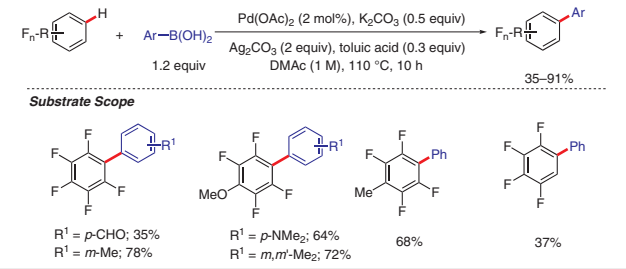
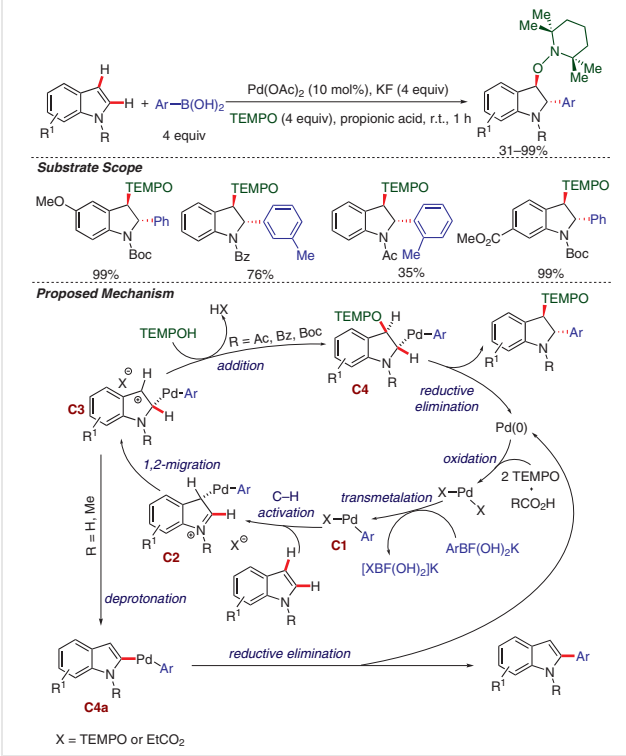


Scheme 4 Direct C–H arylation of electron-deficient polyfluoroarenes through Pd catalysis

Also in 2009, the Studer group reported that various indoles reacted with arylboronic acids chemodivergently. C–H arylation of *N*-unsubstituted indole and *N*-methylindole gave the corresponding C-2-arylated indoles (6 examples 68–81% yield), whereas *N*-Ac, *N*-Bz, and *N*-Boc-protected indoles provided the corresponding arylcarboaminoxylated products with excellent diastereoselectivity in good to excellent yields with the aryl group at the C-2 position and 2,2,6,6-tetramethylpiperidin-*N*-oxy (TEMPO) at C-3 (Scheme 5).[17] Surprisingly, it was found that the only competing reaction was arylation at C-3 as 3-arylindoles were isolated from every mixture of products albeit in minor amounts. The *trans*-configuration of aryl and TEMPO was inferred after performing XRD and HPLC studies of all products. Arguably, the 2,2,6,6-tetramethylpiperidin-*N*-oxy (TEMPO) radical behaved as an oxidant in this reaction. Boc, Bz, and Ac were used as *N*-protecting groups amongst which Boc appeared the most suitable. Arylboronic acids containing F, Br, Me, and OMe and indoles possessing OMe, Me, and Br functionalities were compatible under reaction conditions. Sterically encumbered 2-tolylboronic acid coupled with *N*-substituted indoles to give the corresponding products in moderate yields.

The proposed mechanism (Scheme 5) involved initial transmetalation, followed by C-3 palladation to form **C2**. The slow reaction rate facilitates 1,2-migration of the catalyst and shifts the catalyst to C-2. Channelized by the nature of the protecting group on the indole nitrogen, **C3** either moves to only the arylation pathway (**C4a**) or carbaminoxylation pathway to form **C4**. Both, upon reductive elimination, produce arylated and carbaminoxylated products, respectively. The counteranion could be either $TEMPO^-$ or $EtCO_2^-$. The Ac, Boc, and Bz groups might be involved in Pd catalyst stabilization through C=O–Pd bonding that drastically reduces deprotonation rate, facilitating *trans*-addition of TEMPO to the system. Finally, two equivalents of TEMPO oxidize the catalyst to close the cycle.

In 2010, You and co-workers reported a Pd-Cu co-catalytic system for the direct Suzuki–Miyaura-type C–H arylation of a number of caffeine and other azoles (Scheme 6).[18] The C-8 position of xanthines, which is electrophilic in nature, was arylated with ease by (het)arylboronic acids to produce (het)arylated xanthines that act as selective antagonists at A_{2B} adenosine receptors. Gratifyingly, transmetalation of boronic acids was prevented by 1,4-benzoquinone (BQ), suppressing its homocoupling and easing oxidative coupling with the C–H bond. A wide range of EDGs and EWGs on the arylboronic acid were tolerated, but very few alterations could be made on the caffeine skeleton. Azoles were arylated at C-2 and indolizine was arylated at C-3 using this catalytic system in a single step which required two steps when performed by the traditional Suzuki–Miyaura cross-coupling. Mechanistic studies revealed evidence of trace Cu(I) generation from $Cu(OAc)_2$ that might have driven facile transmetalation.



Scheme 5 Pd-catalyzed C–H arylcarboaminoxylation of *N*-protected indoles and proposed mechanism

In 2011, the Itami group published a report that facilitated efficient arylation of polycyclic aromatic hydrocarbons (PAHs) that could be assembled further to engender shape, width, and topology-tailored graphenes, useful as optoelectronic materials (Scheme 7).[19] A combination of Pd/*o*-chloranil was employed for the arylation of pyrene derivatives, a class of tetracyclic benzenoid PAH, using arylboroxins in 2 h at 80 °C. Notably, phenyl boronic acids and corresponding boronates were compatible but gave low yields. *o*-Chloranil exhibited oxidation aptitude, ascribable

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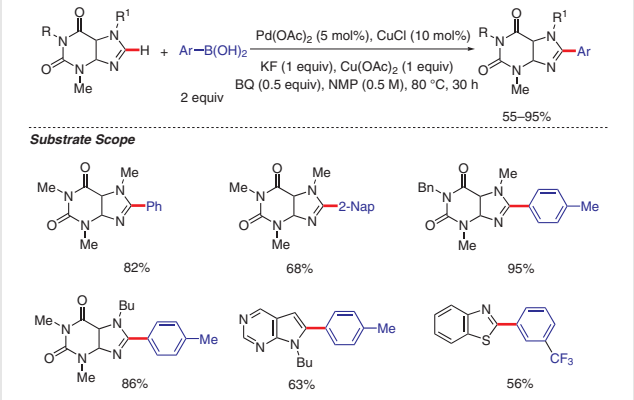
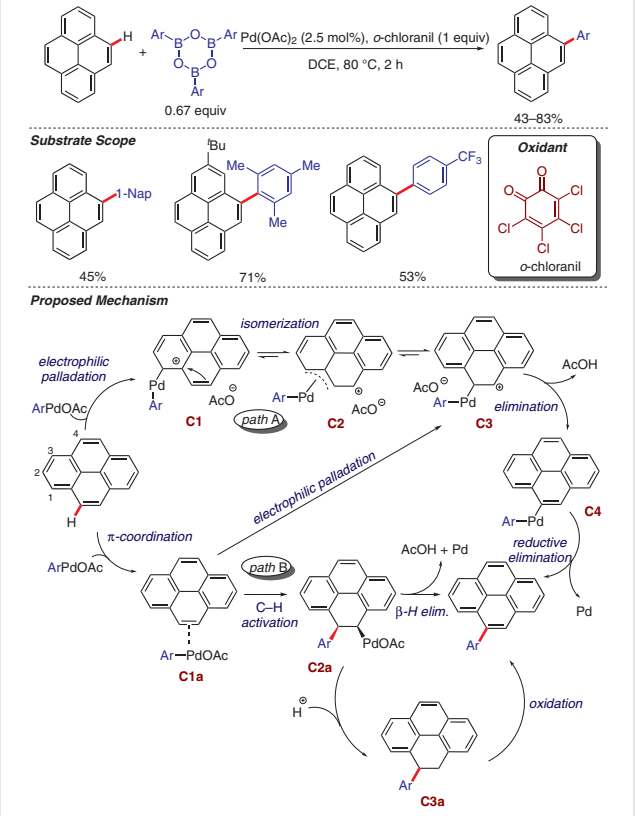


Scheme 6 Substituted caffeine derivative synthesis through Pd-Cu co-catalyzed C–H arylation

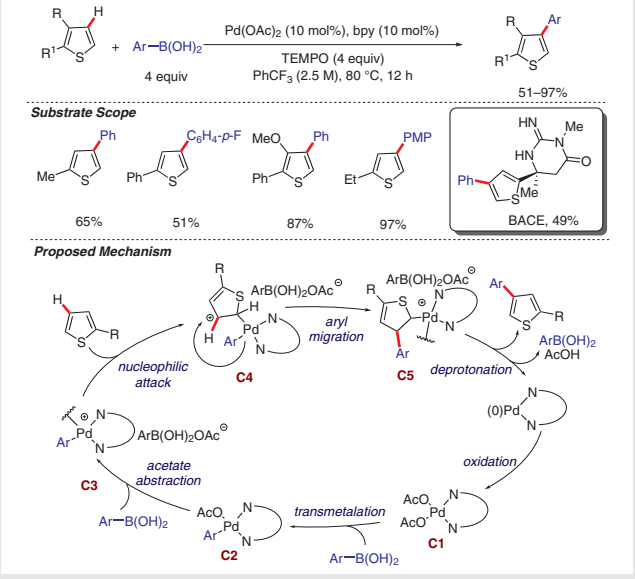
to its *o*-quinone like skeleton and binding ability towards Pd with both oxygens. Sterically hindered *o*-substituted aryls, mesityl and 1-naphthyl were easily introduced at C-4 of pyrenes to give the product in good yields, and this led the Itami group to develop methods[20,21] to synthesize sterically congested biaryl systems. The addition of excess arylboroxin resulted in diarylation at the C-4 and C-5 positions. Use of *o*-biphenylboroxin led to arylation between C-4 and C-5 of 2,7-di-*tert*-butylpyrene, which on oxidation with $FeCl_3$, resulted in a facile expansion of the PAH skeleton through cyclodehydrogenation with 60% yield in gram scale.

Two plausible mechanisms were proposed (Scheme 7). In path A, electrophilic palladation takes place at the most nucleophilic C-1 position and **C1** transforms to **C3** through σ-π-σ isomerization of **C2**. **C3**, which is the most stable intermediate due to an intact pair of aromatic rings, undergoes deprotonation to form **C4**, followed by reductive elimination to produce the product. In path B, pyrene coordinates with the catalyst through π-ligation. If electrophilic palladation occurs, then the stable cationic intermediate **C3** is generated. Otherwise, insertion, like Heck-type carbopalladation, possibly forms **C2a**, which undergoes either protodepalladation/oxidation or common β-elimination to give the desired product.

Thiophene and thiazole arylation at C-4 was accomplished by Studer, Itami, and co-workers using arylboronic acids and the proposed mechanism differed from arylation at either C-2 or C-5 positions (Scheme 8).[22] The high regioselectivity for arylation at C-4, instead of C-5, of 2,3-disubstituted thiophenes was attributed to the addition of excess arylboronic acid. TEMPO only acted as an oxidant and played no role in influencing the exceptional regioselectivity at C-4. Various electron-donating and electron-withdrawing groups on both thiophene and arylboronic acid were accommodated. To illustrate the pharmaceutical utility of this reaction, they synthesized a 4-arylated thiophene which is a potent inhibitor of β-secretase (BACE) and consequently used as a drug in treating Alzheimer's disease.



Scheme 7 Substituted pyrene derivative synthesis through Pd-catalyzed C–H arylation and proposed mechanism



Scheme 8 4-Arylthiophene synthesis through Pd-catalyzed C–H arylation and proposed mechanism

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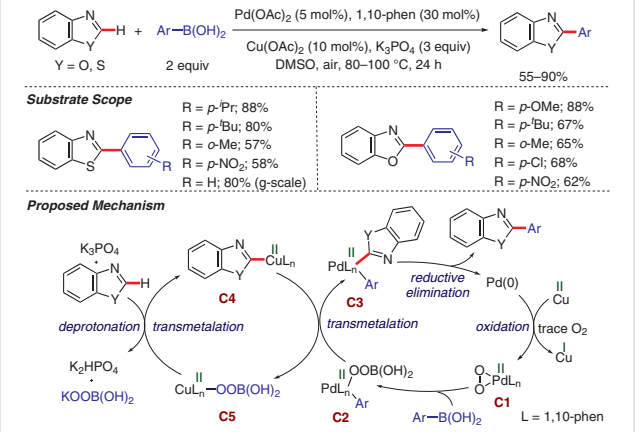
The mechanism is as follows (Scheme 8): upon initial transmetalation, **C2** reacts with another equivalent of arylboronic acid and the acetate anion is abstracted, forming the counteranion $[ArB(OH)_2OAc]^-$, a weaker base compared to the acetate itself. **C3** then undergoes nucleophilic attack by thiophene through C-5 to generate **C4**. It is proposed that the lower basicity of the counteranion slows down the deprotonation step which then effects the aryl migration. In the presence of free acetate, C-5 arylation is the primary reaction pathway but the absence of free acetate results in C-4 arylation. Then, **C5** undergoes deprotonation to produce 4-arylated thiophene and subsequent oxidation by TEMPO brings Pd(0) to Pd(II) that facilitates the catalysis further.

Cyclocondensation is one common method for synthesizing heterocyclic compounds with varied functional groups but its scalability and application in building a library of heterocycles remains difficult due to poor step- and atom-economy. Liu and Ranjit investigated a Pd-Cu cocatalytic system for the arylation of benzothiazoles and benzoxazoles (Scheme 9)[23] that had little in common the conditions used by You and co-workers (see Scheme 6).[18] Reduced catalyst and ligand loading had a small impact on reaction productivity and molecular oxygen from the air acted as a terminal oxidant for the protocol, making the reaction ecofriendly. The arylboronic acid could be substituted with a range functionalities, like Me, OMe, NO_2, *t*-Bu, *i*-Pr, and Ph, and even sterically bulky 2-tolyl- and 1-naphthylboronic acids gave the corresponding products in up to 80% yield. To demonstrate scalability, gram-scale reactions were conducted between benzothiazole and phenylboronic acid that gave 2-phenylbenzothiazole in 80% yield under the optimized conditions.

The proposed mechanism (Scheme 9) is initiated by deprotonation of benzoxazole or benzothiazole and subsequent transmetalation with **C5** to produce heteroarylcuprate **C4**. The intermediate transmetalates with **C2** and **C3** undergoes reductive elimination to produce the desired product and regenerate Pd(0). Trace molecular oxygen oxidizes Pd(0) to form peroxopalladium complex **C1**, in the presence of Cu(II), that transmetalates with $ArB(OH)_2$. If air is replaced by pure oxygen, then the excess oxygen results in oxidative hydroxylation of boronic acid, that competed with its transmetalation, leading to lower efficacy of the catalytic cycle and byproduct formation.

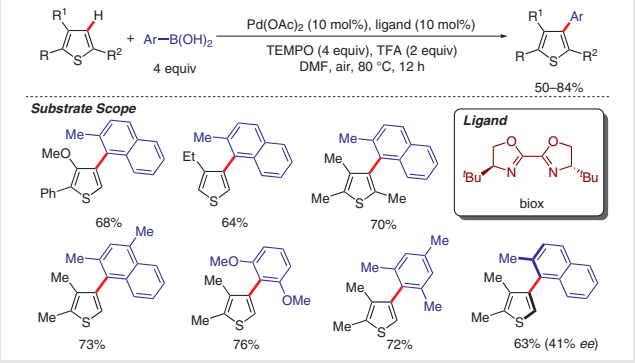
In 2012, the Itami group presented a unique protocol to tackle the arduous challenge of synthesizing sterically encumbered chiral (het)biaryls, circumventing *ortho*-substituent steric issues (Scheme 10).[20] Axial chirality of such molecules makes these useful for application as chiral catalysts, ligands, and bioactive materials, etc. Screening of ligands for use with $Pd(OAc)_2$ identified bisoxazoline (biox) as the most successful and the yield was further improved by the addition of trifluoroacetic acid. 2,3-Disubstituted (benzo)-



Scheme 9 C–H arylation of benzoxazoles and benzothiazoles at C-2 through Pd catalysis and proposed mechanism

thiophenes with Me, OMe, and Et groups coupled with 1-naphthyl-, pyridyl-, and mesitylboronic acids and 2,6-disubstituted arylboronic acids in up to 84% yield. Surprisingly, 2,3,5-trimethylthiophene also gave the arylated product in 58–70% yields, overcoming all possible steric interactions. Applying the optimized conditions, 3-methyl-4-(2-alkylnaphthalen-1-yl)thiophenes with axial chirality were synthesized in up to 63% yield along with moderate ee; experiments showed that their thermodynamic stability to be adequate to remain stable at room temperature. Gratifyingly, indoles and benzofuran also reacted with moderate to good efficiency, thus broadening the substrate scope of the method.

Cyclic enaminones have been observed in the core of medicinally relevant molecules like (–)-antofine, (*R*)-preclamol, nabitan, etc. In 2012, the Georg group developed their previous arylation further (Scheme 1)[8] and established a new set of conditions with a shorter reaction time frame to functionalize C-5 of (bi)cyclic enaminones (Scheme 11).[24] In this approach, the electronic properties of



Scheme 10 Synthesis of sterically congested 4-arylthiophenes by Pd/biox-catalyzed C–H arylation

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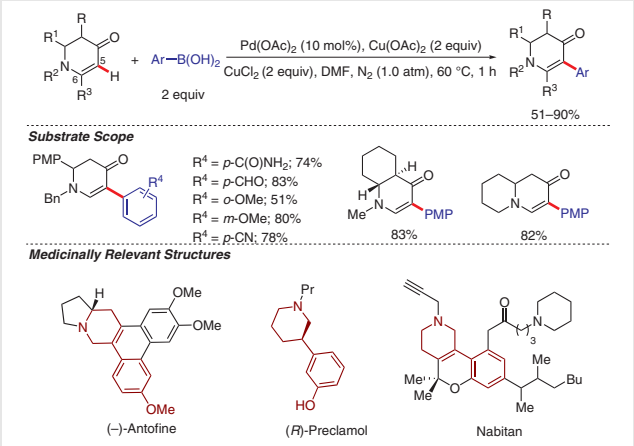
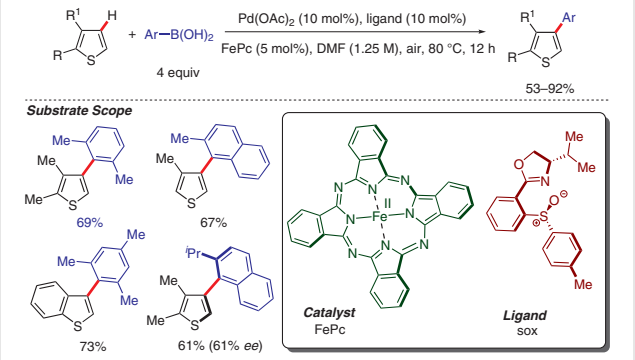
the incoming aryl group on the arylboronic acid do not affect the yields, in contrast to their previous approach[8] that required electron-rich arylboronic acids. Electron-rich ones produce superior results compared to electron-poor ones. Mono- as well as bicyclic enaminones with *N*-protection gave substantial yields when coupled with arylboronic acids bearing halo, CF_3, CHO, CN, and OMe groups; 2-methoxyphenylboronic acid however, gave a low 51% yield. The Bn, Me, Ph, and Cbz groups were utilized as the *N*-protecting group; no regioselective arylation was observed when enamine with free N–H was used. Both $Cu(OAc)_2$ and $CuCl_2$ appear to be necessary as neither of them could facilitate efficient coupling individually and thus, their cooperative oxidative effort was inferred.



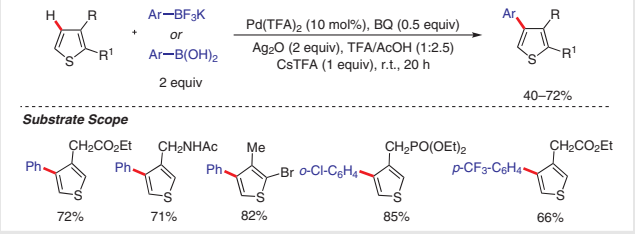
Scheme 11 Pd-catalyzed C–H arylation of (bi)cyclic enaminones at C-5 and relevant pharmacophoric architectures

In 2013, the Itami group reported another strategy for hindered chiral heterobiaryl synthesis that deployed an intricate Pd(II)-sulfoxide-oxazoline/FePc dual catalyst system and postulated the reaction to be ligand enabled (Scheme 12).[21] Presence of FePc was crucial and lowering its loading down to 1 mol% drastically reduced the yield of the product. Initially experiments showed that the use of DMSO as a ligand was required, but optimization identified a hybrid sulfoxide-oxazoline (sox) ligand that improved the yield significantly. Substituted (benzo)thiophenes containing OMe, 2,3-Me_2, 2-Ph on the thiophene ring were arylated with 2,6-dimethylphenyl-, mesityl- and 2-substituted 1-naphthylboronic acids in up to 92% yield. Also, *N*-(un)substituted indoles, furans, and pyrroles reacted albeit indoles generally gave inferior yields compared to thiophenes. As in their previous synthesis (Scheme 10),[20] they also synthesized a chiral heterobiaryl molecule with 61% yield and 61% ee.



Scheme 12 Pd-FePc/sox co-catalyzed C–H arylation of substituted thiophenes at C-4

Keeping thiophene functionalization[25] under focus, Bach and co-workers performed a set of C-4 arylations and speculated on the absence of any chelating impact from the other substituent present at C-2 and C-3 (Scheme 13).[26] When reaction was carried out in acidic medium using electron-poor arylboronic acids then the products of homocoupling were also isolated. Replacing them with potassium aryltrifluoroborate ($ArBF_3K$) improved the yield and also accommodated sterically hindered aryl groups. 2,3-Disubstituted thiophenes provided better yields compared to C-3 substituted thiophenes and halomethyl, alkoxycarbonylmethyl, and phosphorylmethyl groups were tolerated under the optimized conditions.



Scheme 13 Synthesis of 4-arylthiophenes by Pd-catalyzed C–H arylation

In order to synthesize antagonists of *N*-methyl-D-aspartate (NMDA), Messaoudi, Alami, and co-workers needed to synthesize 3-arylquinoxalin-2(1*H*)-ones from quinoxalin-2(1*H*)-ones which they did using the $Pd(OAc)_2$/1,10-phen catalytic system and arylboronic acids (Scheme 14).[27] Arylboronic acids substituted with NH_2, OH, SMe, OMe, Me, Cl, and F and quinoxalin-2(1*H*)-ones substituted at C-6 and/or C-7 with NO_2, F, and Cl were successfully utilized. Quinoxalin-2(1*H*)-ones were mostly protected at N-1 by Me and produced up to 96% yield. Also, Bn, PMB, and CH_2CO_2Et protections were utilized without sacrificing much of that yield.

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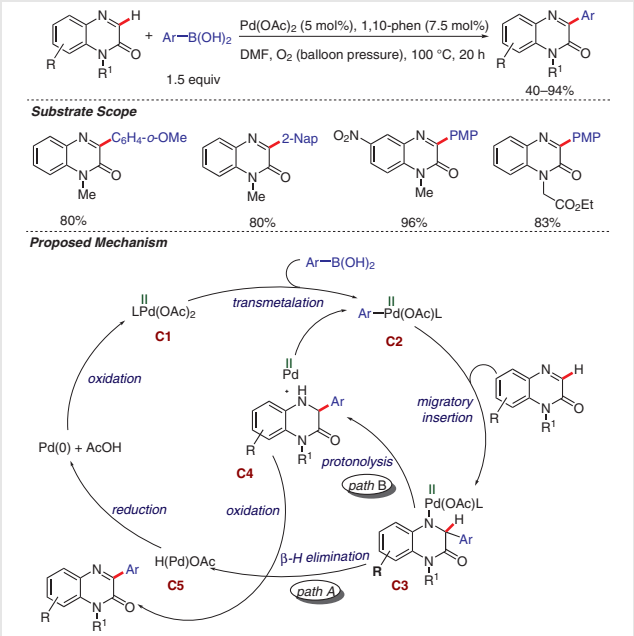
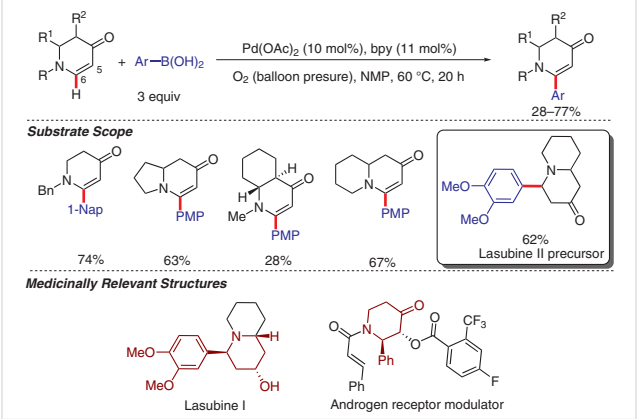
N-1-unsubstituted quinoxalinone reacted with poor yields which was thought to be due to poor solubility. Performing the reaction under an oxygen atmosphere suppressed quinoxalinone hydroarylation. In a subsequent reaction, the 3-arylquinoxalin-2(1*H*)-ones were then subjected to regioselective chelation-assisted *ortho*-functionalization of the newly installed aryl ring.



Scheme 14 Pd-catalyzed C–H arylation of quinoxalinone at C-3 and proposed mechanism

The proposed mechanism (Scheme 14) takes place by initial transmetalation of **C1** followed by migratory insertion of quinoxalinone on **C2** to form **C3**. The intermediate can either undergo protonolysis/oxidation to generate the final product (path B) or transform through simple β-elimination (path A). Finally, oxidation of Pd(0) to Pd(II) by O_2 completes the catalytic cycle. Based upon controlled experiments, it is proposed that path B is more likely than path A as the absence of oxygen mainly produced **C4**, over the Heck-type product.

In 2014, the Georg group extended their enaminone functionalization[8,24] to the arylation of enaminones and, predominantly, bicyclic enaminones at C-6 (Scheme 15).[28] Optimization of the conditions found $Pd(OAc)_2$ with bipyridine (bpy) in NMP with oxygen, as the oxidant, to be the best conditions. They suggested that bidentate ligand coordination and Pd catalyst electron enrichment were key to switch the selectivity from C-5 to C-6. The direct arylation was controlled by the electronic nature of the *N*-protecting group as electron-donating groups (Me, Bn) offered exclusive Heck-type direct C–H arylation, while electron-withdrawing groups (Cbz) also gave conjugate addition products in low yields. Using *N*-Cbz protecting group and 4-methoxyphenylboronic acid, the conditions could be optimized by increasing the acidity, the conjugate addition became the dominant route over direct arylation. Reactions of arylboronic acids containing various functional groups, which could be further synthetically manipulated, were successful, but the reaction of sterically hindered 2-methoxyphenylboronic acid was unsuccessful. To validate biologically relevant applications, they synthesized a lasubine II precursor in 62% yield.



Scheme 15 Pd-catalyzed C–H arylation of (bi)cyclic enaminones at C-6 and relevant pharmacophoric architectures

2.2 Iron Catalysis

In 2013, our group succeeded in the efficient arylation of benzothiazoles, quinoxalines, and thiazoles using a commercially available iron salt, albeit the reaction with other heterocycles like pyrrole and indoles was unsuccessful (Scheme 16).[29] The mild conditions using $Fe(NO_3)_3$, $K_2S_2O_8$, and TFA in THF/H_2O under air were successfully utilized with arylboronic acids containing Ac, *t*-Bu, ester, and halo groups and heterocycles containing CN, Me, and OMe groups. The use of $AgNO_3$ as the catalyst provided only moderate yields but replacing it with $Fe(NO_3)_3$ resulted in improved yields with scalability; the gram-scale reaction gave 2-phenylbenzothiazole in 85% yield. Intriguingly, the scope of the reaction was broadened to the arylation of *p*-quinones, which are widely used ligands in Pd catalysis, to give the corresponding product in substantial yields. This observation implied that the quinones ligated the Fe-catalyst with minimum to no efficiency.

A mechanism was proposed (Scheme 16) based on precedent and mechanistic investigations in which initial homolytic cleavage of persulfate inevitably leads to oxidation of Fe(III) to produce **C2**. Then, the heteroaryl-Fe(IV) intermediate, a well-known oxidation state of Fe, reacts with $ArB(OH)_2$ to effect aryl radical formation that reacts with cationic **C3** to form the desired product.

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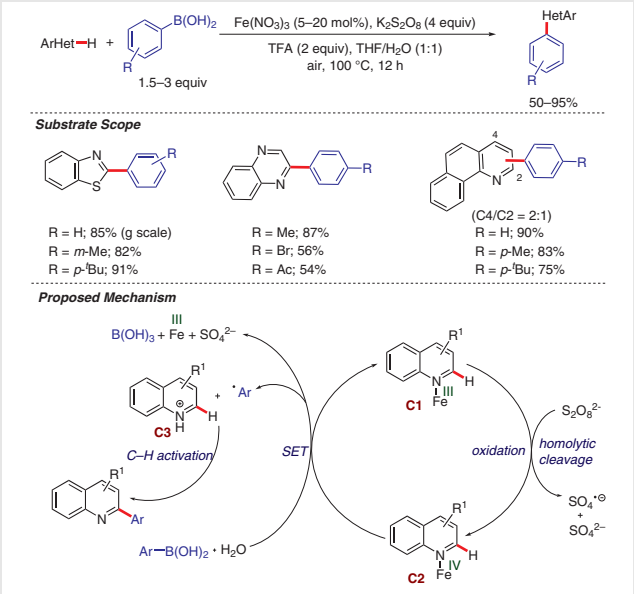
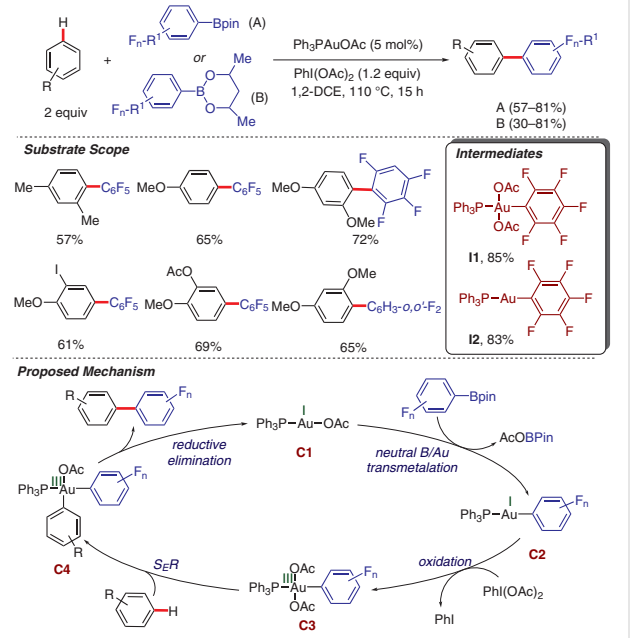


Scheme 16 Fe-catalyzed C–H arylation of benzothiazole, quinoxaline, and thiazoles and proposed mechanism

2.3 Gold Catalysis

Gold, an expensive jewel metal, is not a go-to catalyst for C–H activation or cross-coupling reactions but synthetically inert substrates sometimes require gold catalysts for efficient coupling. In 2017, Nevado and co-workers coupled electron-poor polyfluoroarylboronates and electron-rich arenes in the absence of an external base (Scheme 17).[30] This is the reverse of the work of Su and co-workers (Scheme 4)[16] in which polyfluoroarenes were arylated with electron-rich boronic acids. Under the optimized reaction conditions [$Ph_3PAuOAc$, $PhI(OAc)_2$, 1,2-DCE, 110 °C], arylboronic acids underwent rapid protodeboronation but reactions of arylboronic pinacol esters (ArBpin) were successful in this pseudo-neutral medium where acetate ligand seemed to play the role of an internal base. Electron-rich arenes containing OAc, OMe, OEt, and Me groups gave the corresponding products in satisfactory yields. Two proposed intermediates of the reaction were synthesized in 85% (**I1**) and 83% (**I2**) yields, respectively, and then they were utilized in stoichiometric amounts with arenes to give biaryl product formation in 59% yield thus confirming them as intermediates in the reaction.

According to the proposed mechanism (Scheme 17), the reaction starts with neutral B/Au(I) transmetalation of **C1** and aryl-Au(I) complex **C2** forms. Subsequently, oxidation takes place to provide **C3** having a *trans*-diacetate structure, confirmed via intermediate synthesis. Following the S_ER route, **C4** is formed followed by reductive elimination to regenerate the catalyst and produce the observed product. Notably, in this case oxidation occurs before elimination not, as generally happens, after it.



Scheme 17 Au-catalyzed oxidative C–H cross-coupling between electron-deficient and saturated arenes and proposed mechanism

3 Arylation with Directing Group Assistance

The low number of successful C–H arylations with arylboron compounds of substrates that do not contain a directing group shows that strategic innovation is still required. A large number of studies utilize directing groups for successful regioselective functionalization and directed *ortho*-metalation (DoM), which requires further improvements in the synthesis of substrates containing directing groups and their removal.

3.1 Palladium Catalysis

In 2007, the Shi group reported seminal work on the *ortho*-arylation of arenes exploiting an *N*-alkylacetamide as the directing group (Scheme 18).[31] They squeezed the two-step Suzuki-Miyaura coupling into a single step without any prerequisite for halogenation of acetanilides. $Pd(OAc)_2$ could be replaced by $Pd(dba)_2$ but authors presumed oxidation of Pd(0) to Pd(II), leaving the former as the catalyst of choice. Electron-rich arylboronic acids gave better yields compared to electron-deficient examples; the reaction was sensitive to steric hindrance as 2-methylphenylboronic acid gave the corresponding product in only 20% yield. The arylation of 1-acetyl-2-phenethyl-1,2,3,4-tetrahydroquinoline containing an additional remote Ph group did not result in additional arylation, showcasing the exclusive regioselectivity of the optimized conditions. *N*-Acetyl-2,3-dihydroindoles substituted by Cl and Me groups underwent smooth

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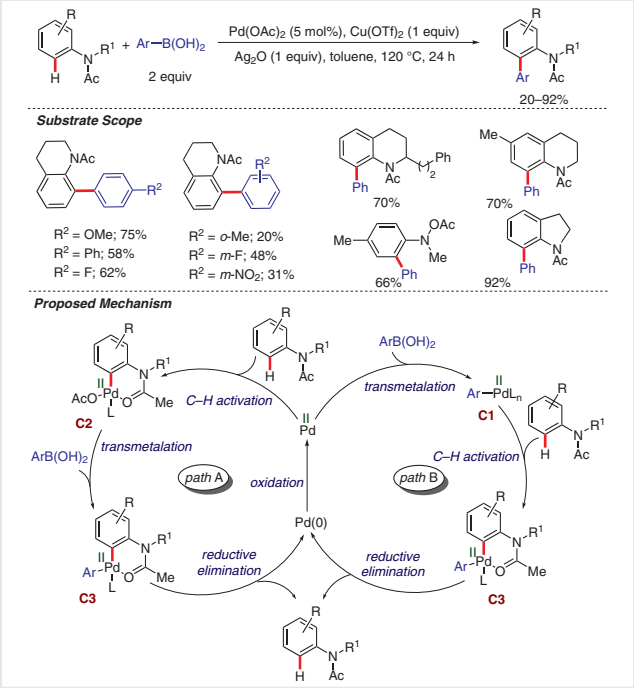
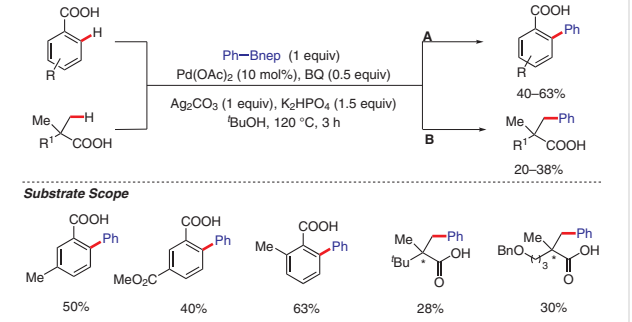
arylation reaction to afford 7-arylated indoles. Kinetic studies revealed a KIE value of 2.3, implying C–H scission in the rate-determining step.



Scheme 18 Pd-catalyzed *ortho*-C–H arylation of arenes directed by *N*-alkylacetamide chelation and proposed mechanism

The proposed mechanism (Scheme 18) has two pathways: path A involves initial C–H activation, followed by transmetalation while path B involves the opposite sequence, but both give intermediate **C3**. Finally, **C3** undergoes reductive elimination to produce the isolated product. Experimentally, the validity of path A was shown as $[(PPh_3)_2PhPdI]_2$, an alternative complex of the probable intermediate **C1**, was subjected to reaction with a cyclic acetanilide which didn't proceed.

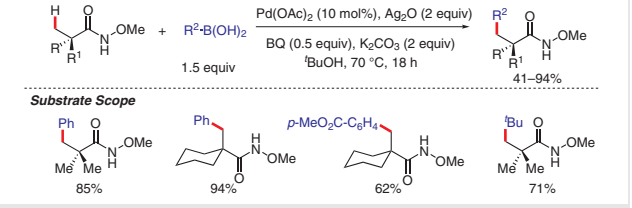
The Yu group published an sp^2 and sp^3 C–H arylation protocol that arylated benzoic acids and aliphatic carboxylic acids at *ortho*- and β-positions, respectively, using arylboronic neopentylglycol ester (ArBnep) (Scheme 19).[32] *ortho*-Arylation was directed by the chelating ability of the carboxylate moiety that resulted through reaction with K_2HPO_4. Ostensibly, *meta*-arylation also took place occasionally but in low yields; benzoic acids substituted with OMe and CO_2Me and aliphatic carboxylic acids substituted with CO_2Me were successfully utilized. In the case of simple aliphatic acid arylation, often diarylation products were obtained in albeit moderate yield, confirming carboxylate as the chelating group. The use of 4-(methoxycarbonyl)-2,2-dimethylbutanoic acid containing a remote ester group on the aliphatic chain illustrates the regioselectivity of the method. To show the diversity of the arylation, sp^3-C–H arylation could be undertaken when arylboronates were replaced by iodoarene and the corresponding products obtained in up to 72% yield.



Scheme 19 *ortho*- and β-C–H arylation of benzoic and aliphatic carboxylic acids through Pd catalysis and carboxy group direction

In 2008, in a breakthrough study by the Yu group developed a method for both arylation and alkylation of sp^3-C–H bonds (Scheme 20).[33] *O*-Methylhydroxamic acids were arylated in the presence of a stoichiometric amount of Ag_2O and benzoquinone (BQ), aided by amide-*N*-chelation. The use of stoichiometric Ag_2O is unfeasible in large-scale syntheses hence they utilized air as a terminal oxidant and the products were obtained in yields comparable to the previous conditions. Hydroxamic acids with aliphatic chains, rings, and remote heteroatomic functionalities were successfully reacted but the arylboronic acid was only phenyl, and 4-CO_2Me- and 4-AcNH-substituted phenyl. While the scope of arylboronic acids is narrow, the ease of transformation of the C(O)NHOMe functionality of hydroxamic acid to esters, amides, and alkanes brings an aspect that expands the synthetic utility of this method.

Also in 2008, the Yu group extended their investigations to the *ortho*-arylation of benzoic acids and arylacetic acids using potassium aryltrifluoroborates with the optimized conditions of $Pd(OAc)_2$ as the catalyst, K_2HPO_4 as the base, *t*-BuOH as the solvent at 100 °C in air for 48 h (Scheme 21).[34] The functionalization was presumed to be directed by the carboxyl group. Potassium aryltrifluoroborates substituted with OMe, Me, and Cl, and benzoic acids substituted with F, Br, I, CN, Ac, and NMe_2 were successfully coupled, but only a few functional groups could be present on the phenylacetic acid (Cl, NO_2, OMe, and alkyl). The product



Scheme 20 sp^3-C–H arylation of *O*-methylhydroxamic acids through Pd catalysis and hydroxamyl group direction

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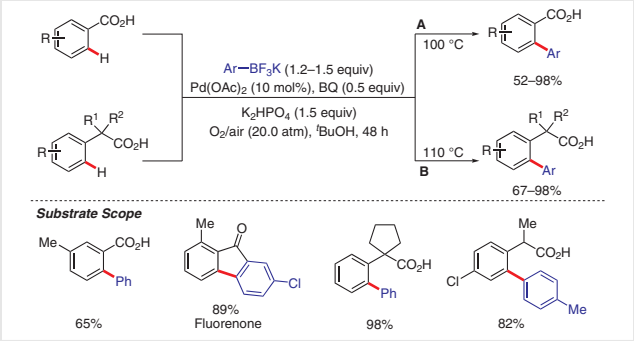
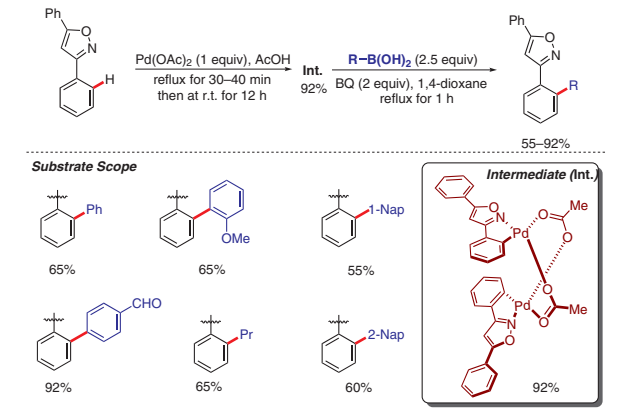
o-arylated acids were in some cases treated with CH_2N_2 and isolated as the methyl esters. In two examples the product *o*-arylated acids were treated with oxalyl chloride to give fluoren-9-ones with 87–89% yields. Also, reaction of benzoic acids with potassium 3-pyridyltrifluoroborate under these conditions gave *o*-3-pyridyl-substituted benzoic acids which subsequently underwent intramolecular annulation to give the tricyclic products in 74–82% yields, thus signaling synthetic relevance of the method.



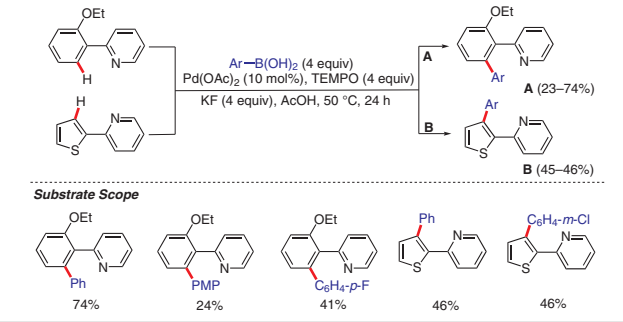
Scheme 21 Aromatic carboxylic acid *ortho*-C–H arylation through Pd catalysis and carboxy group direction

Wu and co-workers (2008) performed the arylation of 3,5-diphenyloxazoles by first reacting 3,5-diphenyloxazole with $Pd(OAc)_2$ to give an isoxazole palladacycle complex that was reacted with arylboronic acids using 1,4-benzoquinone as the co-oxidant to give 3-(2-substituted phenyl)-5-phenylisoxazoles (Scheme 22).[35] This two-step method exhibited tolerance to a bunch of functionalities, like NO_2, 1-naphthyl, CHO on $ArB(OH)_2$ albeit selectively preferring electron-rich boronic acids over electron-poor ones. Interestingly, alkylboronic acids were also used successfully affording 3-(2-alkylphenyl)-5-phenylisoxazoles in up to 65% yield, the alkyl group could be a cycloalkane, which expands the substrate scope for the method.

The Studer group utilized a pyridyl anchor for the direct *ortho*-arylation of 2-arylpyridines with arylboronic acids using $Pd(OAc)_2$ as the catalyst in the presence of commercially available TEMPO as an oxidant (Scheme 23).[36] The substitution on the arylboronic acid was F, Cl, Me, OMe, and vinyl groups but the corresponding products were produced in only moderate to good yields without any recognizable trend. Heteroarylboronic acids containing 2-thienyl and 2-furyl moieties did not react. Arylation of *o*-substituted 2-arylpyridines gave the arylation products in up to 74% yield albeit with limited substrate scope. Following their primary interest in thiophene chemistry, 2-thienylpyridines were arylated at C-3, however the yields were moderate (46%).

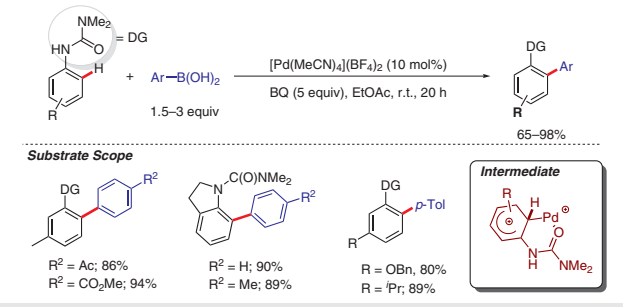


Scheme 22 Pd-catalyzed *ortho*-C–H arylation of arenes through oxazolyl group chelation



Scheme 23 Pd-catalyzed C–H arylation of arenes and thiophenes through pyridyl group direction

In 2010, Lipshutz and co-workers used urea as a directing group for an exclusive monoarylation of arenes with arylboronic acids by a Suzuki–Miyaura-type coupling (Scheme 24).[37] Neutral Pd catalysts often require high temperature to carry out C–H activation however, a cationic Pd catalyst was deployed in this instance that allowed a room temperature reaction. Although, neutral salt $Pd(OAc)_2$ could



Scheme 24 *ortho*-C–H arylation of substituted arenes through Pd catalysis and assisted by urea chelation

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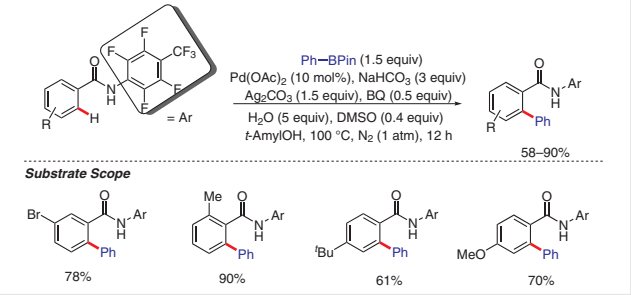
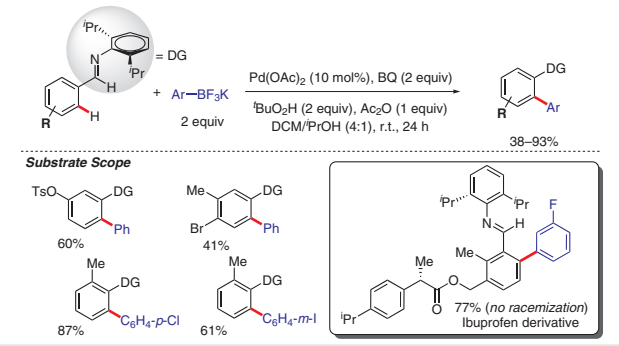
be used in the presence of a strong acid such as HBF_4, $[Pd(MeCN)_4](BF_4)_2$ was more efficacious in generating the positively charged catalyst. A wide range of functionalities, like OMe, CO_2Me, Ac, *s*-Bu, Me, and Cl, were accommodated. Electron-rich arene and electron-deficient arylboronic acid pair engendered excellent yield of the corresponding products. Surprisingly, sterically hindered 2,5-dimethoxyphenylboronic acid reacted with 1,1-dimethyl 3-(3-tolyl)urea to afford the 2-arylated product in 98% yield, thus showing the elegance of the method.

In 2011, the Yu group performed the *ortho*-phenylation of benzamides using phenylboronic pinacol ester and $Pd(OAc)_2$ catalyst, Ag_2CO_3, $NaHCO_3$, and 1,4-benzoquinone in DMSO/water as solvent (Scheme 25).[38] Interestingly, the addition of water appeared crucial to improve the product yield by up to 20% by facilitating the transmetalation of the PhBpin. Various functionalities were tolerated on the benzamides and the biphenylcarboxamides were obtained in up to 90% yield. A critical problem was the occurrence of diphenylation at both *ortho*-positions of the benzamide, but the 2,6-diphenylbenzamides were produced in trivial yield. Vinylation and alkylation of benzamides were also carried out with moderate success, expanding the scope of the method. Illustrating the synthetic utility of the method, 3-methylbiphenyl-2-carboxamide was hydrolyzed to 3-methylbiphenyl-2-carboxylic acid in 90% yield.



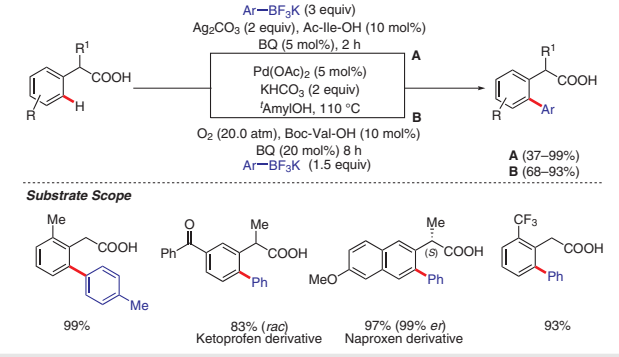
Scheme 25 *ortho*-C–H arylation of substituted arenes through Pd catalysis and assisted by amide group chelation

Carbonyls are a synthetically prominent class of compounds and the functionalization of carbonyl-containing feedstock is highly desirable. However, the weak directing ability of the carbonyl groups is a barrier to such reactions. The Gaunt group used imines as a temporary directing group for the *ortho*-arylation of benzaldehydes at room temperature (Scheme 26).[39] Imines are prone to hydrolysis to give amines that promptly trap the catalyst, thus, the imine of benzaldimine was substituted by a bulky group, 2,6-diisopropylphenyl, to prevent it from degrading and forming a stable polynuclear cyclic palladacycle in the presence of $Pd(OAc)_2$. Thus treating *N*-2,6-diisopropylphenylbenzaldimines with potassium aryltrifluoroborates in the presence of $Pd(OAc)_2$ as the catalyst, 1,4-benzoquinone as an oxidant, *tert*-butyl hydroperoxide, and acetic acid gave the corresponding 2-arylated benzaldimines in up to 93% yield. The benzaldimines were substituted by Me, Ph, F, Br, CH_2OBn, $SiMe_3$, OTs, CO_2Me, NO_2 and while the potassium aryltrifluoroborates were phenyl substituted by Me, halo, Ac and naphthyl groups. Notably, an enantiomer of an ibuprofen derivative functionalized as the imine underwent arylation with potassium 3-fluorophenyltrifluoroborate to give the corresponding *ortho*-arylation product in 77% yield without any racemization. This illustrates the broad substrate scope as well as pharmaceutical relevance of the developed conditions.



Scheme 26 *ortho*-C–H arylation of arenes through Pd catalysis and assisted by imine group chelation

Also in 2011, the Yu group disclosed two similar methods for the ligand-accelerated C–H arylation of phenylacetic acids containing electron-withdrawing and electron-withdrawing groups (Scheme 27).[40] Monoprotected amino acids (MPAAs) emerged as potent ligands for palladium catalysis and remained compatible with all steps of the activation pathway. Potassium aryltrifluoroborates and arylboronic pinacol esters exhibited comparable efficiency in the presence of the $Pd(OAc)_2/Ag_2CO_3$/1,4-benzoquinone/$KHCO_3$/*t*-amylOH system using Ac-Ile-OH as the ligand whereas $ArB(OH)_2$ remained ineffective. Examination of the reaction rate profile showed the efficiency comparability as slow hydrolysis of $ArBF_3K$ produced intermediate $ArB(OH)_2$ in situ.



Scheme 27 Pd-catalyzed *ortho*-C–H arylation of aromatic acids using Ag_2CO_3 or O_2 as oxidants and assisted by carboxyl group direction

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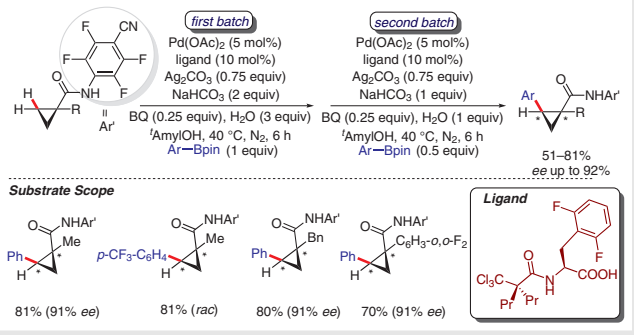
In the first protocol, 1,4-benzoquinone and stoichiometric Ag_2CO_3 were exploited as oxidants in the $Pd(OAc)_2$/Ac-Ile-OH/$KHCO_3$/*t*-AmylOH system for the arylation of phenylacetic acids with potassium aryltrifluoroborates at 110 °C for 2 h. The second protocol comparable yields were obtained by replacing Ag_2CO_3 with oxygen and using Boc-Val-OH as the MPAA ligand, but the use of 1,4-benzoquinone was still required. Phenylacetic acids with alkyl, alkoxy, and halide groups were compatible and α-monosubstitution did not impact on reactivity. Potassium *o*-tolyltrifluoroborate was a poor arylating agent for steric reasons. Nonsteroidal anti-inflammatory drugs (NSAIDs) like ketoprofen and naproxen produced arylated products in 83% and 97% yields, demonstrating late-stage functionalization utility. That the reaction could be carried out under air holds promise for industrial applications. Control experiments suggested a concerted metalation-deprotonation (CMD) pathway in the presence of Ac-Ile-OH and an electrophilic palladation pathway in the absence of the ligand.

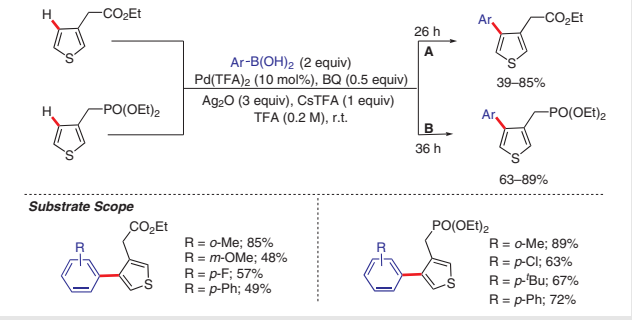
Bach and co-workers reported the arylation of 3-substituted thiophenes with arylboronic acids at the inert C-4 position, catalyzed by the $Pd(TFA)_2$/1,4-benzoquinone/Ag_2O/CsTFA system at room temperature (Scheme 28).[25] Electron-poor arylboronic acids reacted slowly and exhibited poor reactivity under the applied conditions in contrast to electron-rich, sterically encumbered ones. Both 2-tolyl- and 1-naphthylboronic acids gave the corresponding products in excellent (85%) yields, however the reaction with mesitylboronic acid was unsuccessful. 3-Thienylacetates and 3-thienylmethylphosphonates provided the 4-arylated products in good to excellent yields and they could be subjected to further transformations. It is proposed that the functionalities of 3-thienylacetates and 3-thienylmethylphosphonates activate or direct the reaction as the reaction of 3-methylthiophene was slow.

The Yu group reported a stereoselective arylation protocol for the oxidative coupling of the sp^3-C–H of prochiral sterically encumbered, all carbon-quaternary center containing cyclopropanes (Scheme 29).[41] Arylboronic pinacol esters and $NaHCO_3$ were crucial for arylation, but changing the organoboron to potassium aryltrifluoroborates also promoted the reaction. The protocol using $Pd(OAc)_2$/ligand/Ag_2CO_3/$NaHCO_3$/1,4-benzoquinone/H_2O/*t*-AmylOH was optimized to operate at even 40 °C and gave enantioselective arylation products in good to excellent yields and with excellent enantiomeric excess. Increasing the temperature reduced the enantiomeric excess, however the vinylation reaction using vinylboronic pinacol esters could be performed at 100 °C with reasonable enantiomeric excess. Monoprotected amino acids (MPAA) were again the ligand of choice[40] and an aryl ring on ligand backbone was necessary to sustain impressive enantiomeric excess outcomes. In the enantioselective examples the cyclopropyl ring was substituted with R = Me, Pr, *i*-Bu, *i*-Pr, Bn, $(CH_2)_2Ph$, cyclopentyl, CH_2OBn, $(CH_2)_2NPhth$, 4-ClC_6H_4, 2,6-$F_2C_6H_3$, and phenylboronic pinacol ester were successfully utilized; 3-fluoro-4-(trifluoromethyl)phenylboronic pinacol ester was successfully used in the reaction. Notably, a requirement of the reaction is that the addition of arylboronic pinacol ester and the catalytic reagents is made in batches, as their addition all at once produced inferior yields.



Scheme 29 sp^3-C–H arylation of prochiral cyclopropanes through ligand-enabled Pd catalysis and assisted by amide group chelation

Arrayás, Fernández-Ibáñez, Carretero, and co-workers reported the synthesis of biaryls from (2-pyridylsulfinyl)arenes, containing a 2-pyridyl sulfoxide chelating group, and arylboronic neopentyl esters (ArBnep) in the presence of $Pd(OAc)_2$ (Scheme 30).[42] Phenylboronic acid underwent homocoupling faster compared to oxidative cross-coupling with the C–H bond. Methylboronic neopentyl ester was used for the *ortho*-alkylation of a number of (2-pyridylsulfinyl)arenes. Arylboronic neopentyl esters containing electron-donating groups (Me and OMe) and electron-withdrawing groups (Cl and F) were used successfully. Thorough investigations revealed incomplete arylation of arenes and 50–80% effective transformation was realized. The isolation of a palladacycle and its X-ray crystallographic study validated the chelation of the directing group. The directing group was removed by using *n*-BuLi at low temperatures to produce the corresponding arenes in up to 91% yield.



Scheme 28 Pd-catalyzed C–H arylation of 3-substituted thiophenes at C-4 through ester and phosphonate group direction

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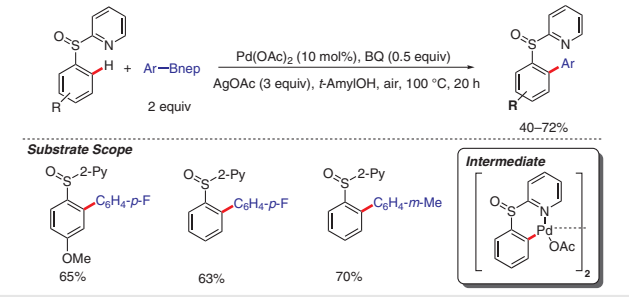
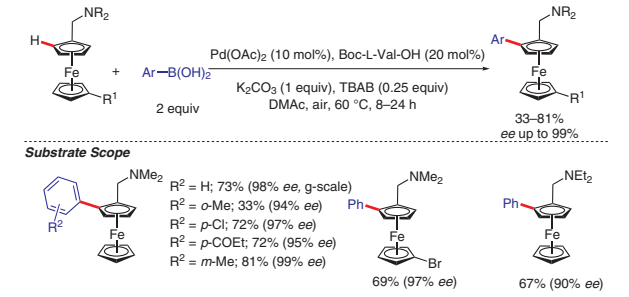


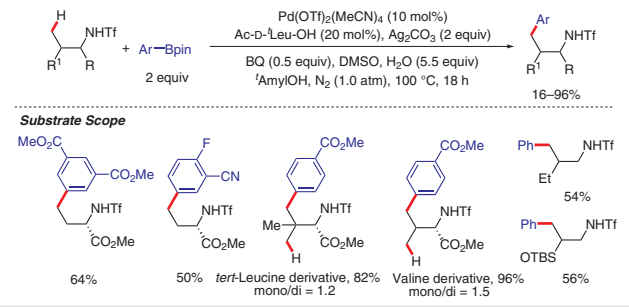
Scheme 30 Pd-catalyzed *ortho*-C–H arylation of substituted arenes directed by 2-pyridyl sulfoxide group chelation



Scheme 31 Pd-catalyzed ligand-accelerated C–H arylation of substituted ferrocenes through amine group chelation

You, Gu, and co-workers synthesized planar chiral ferrocene complex by means of catalytically enantioselective arylation using MPAA ligands (Scheme 31).[43] Boc-L-Val-OH was the ligand of choice for the arylation of aminomethyl-substituted ferrocenes due to its ready availability and high enantiomeric excesses obtained. The ligand choice had a crucial role that might be more important than the directing abilities of the amine functionality. The method uses $Pd(OAc)_2$, Boc-L-Val-OH, K_2CO_3, and TBAB in *N*,*N*-dimethylacetamide (DMAc), in air at 60 °C and gave the products in up to 81% yield and up to 99% ee. The arylboronic acid had substituents such as F, Cl, OMe, Me, CF_3, and CO_2Et, but only unsubstituted chiral (aminomethyl)ferrocenes containing various aminomethyl groups and a bromo substituents (4 examples) were successfully utilized. To demonstrate the scalability of the reaction, 1-[(dimethylamino)methyl]-2-phenylferrocene was synthesized in 73% yield with 98% ee on a gram scale.

In 2014, the Yu group reported a brilliant protocol for the palladium-catalyzed arylation of the γ-C–H bond of triflyl-protected amines and amino acids using arylboronic pinacol esters enabled by MPAA (Scheme 32).[44] Gratifyingly, no competing reaction could be traced in the absence of the MPAA as the ligands provided bidentate coordination assistance to the catalyst to boost arylation outcomes. Bulky and sterically obstructive side chains on the MPAA backbone improved the yield and Ac-L-Leu-OH appeared the most suitable ligand due to easy availability. L-MPAAs produced better yields in D-amino acid arylation and vice versa, implying the spatial influence of ligands on the catalytic cycle. Amino acid substrates with long aliphatic chains, ester groups, amine substrates with TBSO, remote OBn groups, and arylboronic pinacol esters with CO_2Me, F, Cl, Br, OMe, NHAc, and CF_3 were well tolerated under the optimized conditions but the use of heteroarylboronic pinacol esters was not successful. HPLC results revealed no significant racemization at the chiral center present in the substrate and arylation at β-C–H of an amino acid was observed in one instance. Occasionally, diarylation took place albeit in minor amounts. To demonstrate the utility of the method, a library of nonprotein amino acids, valine and *tert*-leucine derivatives were synthesized in 96% and 82% yields, respectively.



Scheme 32 Pd-catalyzed ligand-enabled sp^3-C–H arylation of amines and amino acids assisted by amine group direction

The Yu group then reported work similar to that in Scheme [29][41] by the chiral mono-*N*-protected α-amino-*O*-methylhydroxamic acid (MPAHA) ligand activation of the sp^3-C–H of prochiral cyclobutane systems with arylboronic pinacol esters that gave a sterically congested chiral all-carbon quaternary center bearing the cyclobutane system ligand (Scheme 33).[45] Using a tailored MPAHA ligand under palladium catalysis gave excellent enantioselectivity of up to 95% ee. MPAHA ligands containing an aryl ring gave exceptional ee while *N*-protection of the α-amino group as its carbamate was crucial to sustaining the enantiomeric excess; the parent α-amino-*O*-methylhydroxamic acid (MPAA) produced inferior yield along with deteriorating enantiomeric excess. Heteroatoms were tolerated on the exocyclic alkyl chain and the aryl group of the arylboronic pinacol esters could contain various useful moieties. The arylation of the β-C–H bonds of acyclic geminal dimethyl substrates was examined and the products were obtained with moderate yields.

Chiral α-functionalized amines are the core architecture of various drug molecules that are potential receptor antagonists and inhibitors. In 2016, the Yu group demonstrated

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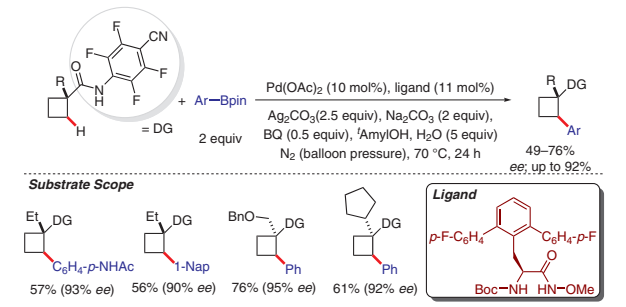
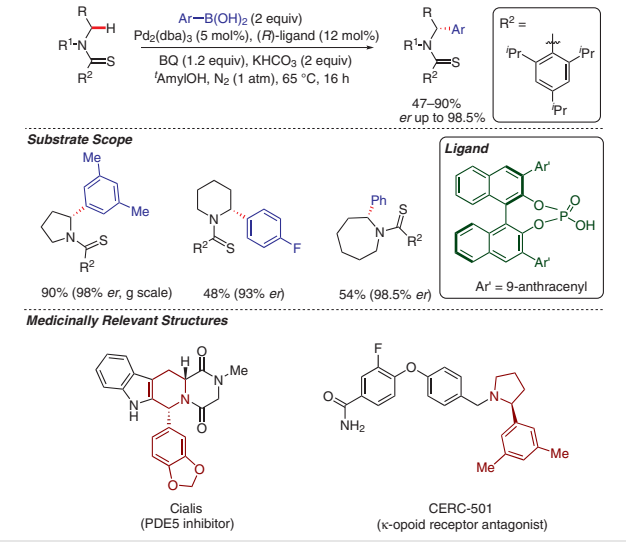


Scheme 33 sp^3-C–H arylation of prochiral cyclobutanes through ligand-enabled Pd catalysis and assisted by amide group chelation

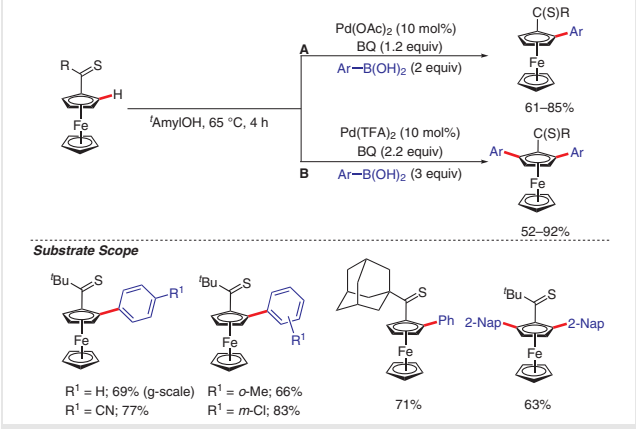
the palladium-catalyzed stereoselective and regioselective α-arylation of cyclic thioamides in the presence of chiral binaphthol-derived ligands (Scheme 34).[46] The investigation began with the use of MPAA ligands but this route was not productive. Next, the use of binaphthol-based phosphoric acid ligands was examined and this gave an excellent level of stereocontrol. The reaction took place under mild reaction conditions and tolerated arylboronic acids containing synthetically valuable functionalities, like CN, CHO, Ac, CO_2Me, Cl, and F. Surprisingly, acyclic thioamides also underwent the reaction albeit with only moderate yields. The optimized protocol using $Pd_2(dba)_3$, binaphthol-based phosphoric acid ligand, 1,4-benzoquinone, $KHCO_3$ in *t*-AmylOH under nitrogen at 65 °C gave the arylated products in up to 90% yield and 98.5% er for the cyclic thioamide systems. The reaction was performed on a gram-scale for a cyclic thioamide and gave the arylated product in 90% yield and 98.5% er.



Scheme 34 Pd-catalyzed α-C–H arylation of (a)cyclic amines through thioamide group assistance

You and co-workers reported the arylation of thiocarbonyl-substituted ferrocenes with arylboronic acids; the thioketone holds significant ligation capability towards Pd catalyst and its subsequent poisoning (Scheme 35).[47] Interestingly, thioketone acted as an anchor in directing the arylation as unlike in a previous study (Scheme 31),[43] no external ligand was required to facilitate the reaction. The use of $Pd(OAc)_2$ as the catalyst with 1.2 equiv of 1,4-benzoquinone resulted in exclusive monoarylation product while the use of $Pd(TFA)_2$ with 2.2 equiv of 1,4-benzoquinone gave predominantly the diarylated product. The method appeared to be unaffected by steric factors as 2-tolyl- and 1-naphthylboronic acids gave the respective products in 66% and 61% yields. Notably, the thiocarbonyl could be substituted with *tert*-butyl, *tert*-pentyl, or 1-adamantyl, but the use of methyl- or phenylthiocarbonyl groups was unsuccessful. The protocol tolerated many synthetically relevant functionalities and the thiocarbonyl group could be transformed into carbonyl by using Ag_2CO_3 at room temperature in 96% yield. A gram-scale arylation of (*tert*-butylthiocarbonyl)ferrocene with phenylboronic acid gave the corresponding product in 69% yield.

In 2017, Lee and co-workers reported the *O*-methyloxime-assisted Pd-catalyzed *ortho*-arylation of acetophenone *O*-methyloximes using boronic pinacol esters (Scheme 36).[48] The presence of electron-donating groups on the boronic pinacol esters resulted in extensive homocoupling, but electron-withdrawing groups improved the productivity. The use of Ac-Gly-OH as an efficient ligand with $Pd(dba)_2$ precatalyst gave the arylated products in up to 80% yield. To simplifying the coupling of disubstituted boronic pinacol esters and acetophenone *O*-methyloximes, first a disubstituted arene was borylated under Ir catalysis[13] and the crude disubstituted arylboronic pinacol esters was subsequently used in the arylation of acetophenone *O*-methyloxime to give the arylated oxime in 89% yield, which was subjected to hydrolysis to give the ketone in 59% yield.



Scheme 35 Pd-catalyzed C–H arylation and diarylation of substituted ferrocenes through thioketone group chelation

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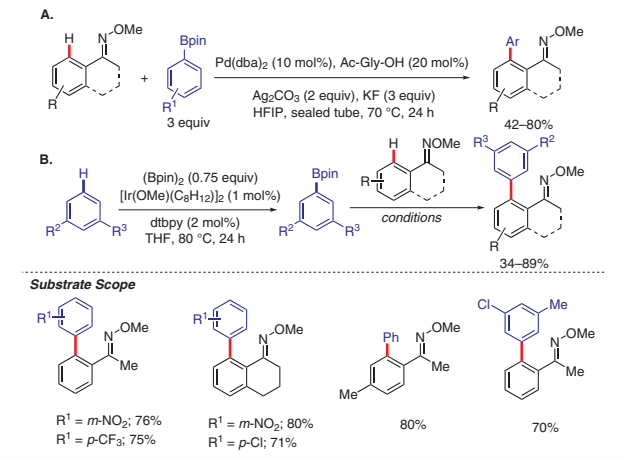
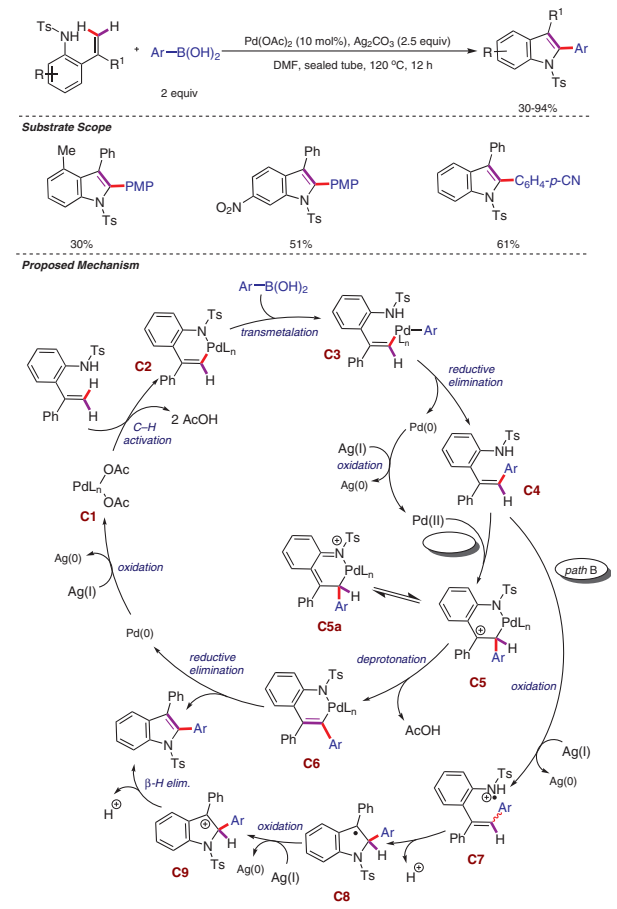


Scheme 36 *ortho*-Arylacetophenone synthesis through ligand enabled Pd-catalyzed C–H arylation assisted by *O*-methyloxime group chelation

In 2018, Zeng and co-workers reported a tandem arylation-amination strategy that could directly access 2-arylindoles from 2-vinylanilines and arylboronic acids (Scheme 37).[49] The 2-vinylanilines were protected by the tosyl group which improved the yield but the yields were unimproved by acetyl or mesityl protection. 2-Vinylanilines substituted by Me, OMe, and Cl groups gave the corresponding *N*-tosylindoles in up to 94% yield but a 5-nitro-2-vinylaniline containing the highly electron-demanding NO_2 group gave the corresponding indole in only 51% yield. The *ortho*-methyl in 3-methyl-2-vinylaniline had a steric impact on the reaction and the indole was obtained in only 30% yield. Surprisingly, 2,6-dimethoxyphenylboronic acid gave the arylated indole in 83% yield. Control experiments showed that the addition of TEMPO did not affect the coupling yield however the addition of 3,5-di-*tert*-butyl-4-hydroxytoluene (BHT) reduced it severely, indicated a possible free radical mechanism. It was inferred that Ag_2CO_3 acts as the terminal oxidant and promoter of the amination pathway through free radical formation.

The proposed mechanism (Scheme 37) involves C–H activation and Ag-promoted single electron transfer (SET) driven ring closure. Initially, the catalyst **C1** is attacked by aniline and the acetate anions are lost and the vinylaniline undergoes vinylic C–H activation to give **C2**. Transmetalation of **C2** with the arylboronic acid gives **C3** which undergoes reductive elimination to give Pd(0) and **C4**. The reaction then proceeds in two possible ways. In path A, oxidized Pd(II) may be attacked electrophilically by the vinylic system to form **C5** and subsequent base-assisted deprotonation to produce **C6**. Upon reductive elimination, **C6** yields the substituted indole. In path B, Ag(I) oxidizes **C4** to form a radical cation **C7** that proceeds to radical **C8** through intramolecular annulation. Oxidation of **C8** by Ag(I) gives **C9** then deprotonation produces the isolated product in good yields.



Scheme 37 2-Arylindole synthesis through Pd-catalyzed C–H tandem arylation-amination assisted by amine group ligation and proposed mechanism

In an extension of the work of You, Gu, and co-workers (Scheme 31),[43] in 2019 Sebesta and co-workers investigated the diastereoselective synthesis of axially chiral (1-aminoethyl)ferrocenes exploiting a commercially available MPAA (Scheme 38).[50] The best results were obtained with $Pd(OAc)_2$ as the catalyst, (*R*)-Boc-Ala-OH as the ligand with K_2CO_3 and TBAB in DMAc in air at 60 °C. Using this procedure, (*R*)-(1-aminoethyl)ferrocenes were arylated using (*R*)-Boc-Ala-OH as the ligand to give the (R,R_p)-arylferrocene product whereas the use of (*S*)-Boc-Ala-OH resulted in lower yields due to spatial influence[44] of the ligand. Likewise, for the arylation of (*S*)-(2-aminoethyl)ferrocenes the best yields were obtained with (*S*)-Boc-Ala-OH and lower yields with (*R*)-Boc-Ala-OH. The particular spatial orientation of (*R*)-Boc-Ala-OH was found to be best and arylboronic acids, or in some case arylboronic pinacol esters, containing Me, OMe, *t*-Bu, Cl, F, and CN were successfully utilized. The reaction was only successful for [1-(dimethylamino)ethyl]ferrocenes, the use of other groups on the amine lead to lower yields due to substrate decomposition.

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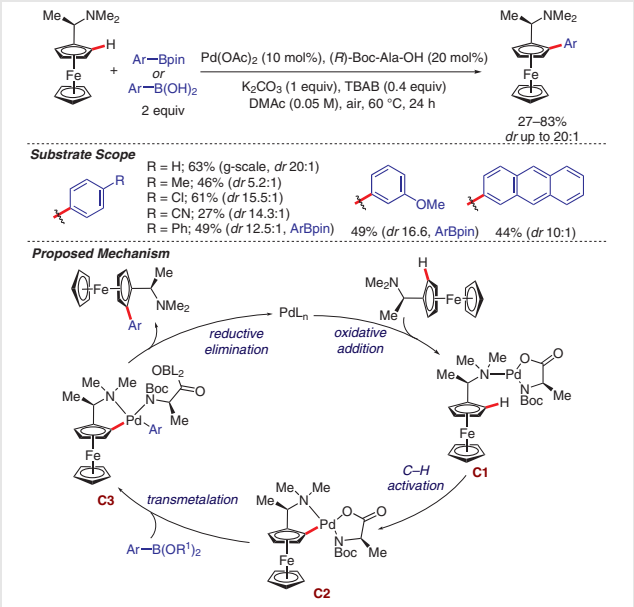


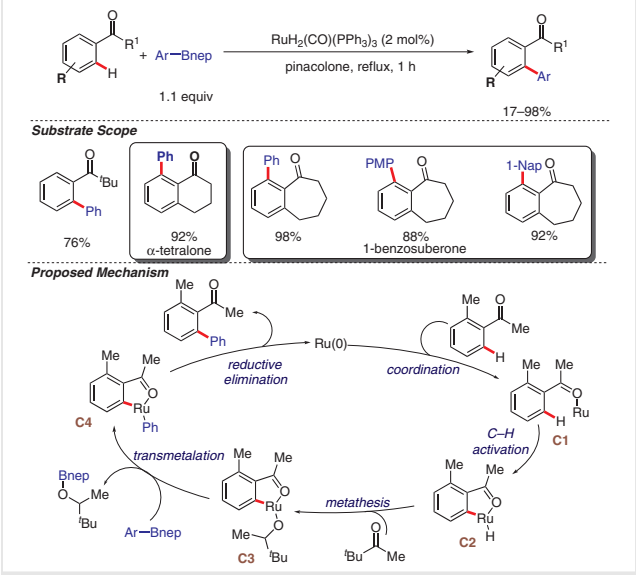
Scheme 38 Pd-catalyzed C–H arylation of substituted ferrocene directed by amine group ligation and proposed mechanism

A straightforward mechanism (Scheme 38) was proposed that demonstrated directing effect of the amine group. First chelation of the amine nitrogen to Pd occurs to form **C1**, and this is followed by a C–H activation step giving **C2**, which undergoes transmetalation giving **C3**. Finally, **C3** undergoes reductive elimination to give the product and regenerate the catalyst.

3.2 Ruthenium Catalysis

The use of arylboronic acids in the presence of ruthenium has often been problematic due to homocoupling or decomposition issues.[51] Therefore, the free OH groups of arylboronic acids are masked as the boronate esters and these are utilized as the arylating agent.

In 2005, Kakiuchi and co-workers reported a breakthrough study for the arylation of aryl ketones using $RuH_2(CO)(PPh_3)_3$ as the catalyst and arylboronic neopentylglycol esters as the arylating agent (Scheme 39).[52] Upon thorough investigation of the predicted mechanism, carbonyl chelation appeared more prominent compared to aromatic ring ligation, prompting the conclusion of carbonyl anchor driving the arylation. An unwanted reduction reaction[53–57] was also encountered when the reaction was performed in toluene, the reduction of the aryl ketone to an aryl alcohol, and this suggests reduction by the Ru–H bond; this reaction was suppressed by using pinacolone or acetone as a sacrificial hydride scavenger, with pinacolone being the preferred scavenger. Aryl ketones containing electron-withdrawing groups (CF_3 and F) and electron-donating groups (Me and OMe) underwent smooth arylation to give the corresponding products in up to 98% yield. 2-Tolyl- and 1-naphthylboronic neopentylglycol esters reacted with 6,7,8,9-tetrahydrobenzocyclohepten-5-one to give the arylation products in 96% and 92% yields, respectively, which indicates that steric factors were not important. Moreover, fused ketones like α-tetralone and 6,7,8,9-tetrahydrobenzocyclohepten-5-one also reacted well under the optimized conditions.



Scheme 39 Ru-catalyzed *ortho*-C–H arylation of aromatic ketones assisted by ketone group chelation and proposed mechanism

The proposed mechanism based on experimental evidence is shown in Scheme 39. First Ru is chelated by the carbonyl group giving **C1**, and it results in *ortho*-C–H activation giving ruthanacycle **C2**. Then, the pinacolone carbonyl ligates with the metal and abstracts the Ru–H bond, thus behaving as a sacrificial hydride scavenger, forming **C3**. The alkoxy-ruthenium intermediate **C3** is transmetalated by the arylboronate with loss of a borate. Finally, reductive elimination of **C4** generates the target product and regenerates the active catalyst.

The functionalization of an inert sp^3-C–H bond is difficult, but the presence of a heteroatom nearby eases the process to a certain extent. In 2006, the Sames group arylated the sp^3-C–H bonds of pyrrolidine and piperidine that are susceptible to β-hydride elimination (Scheme 40).[58] Pinacolone was also used as sacrificial hydride scavenger for M–H, similar to the method of Kakiuchi and co-workers.[52] 2-Substituted *N*-(1-pyrrolin-2-yl)pyrrolidines, containing 1-pyrroline as an efficient directing group, reacted with arylboronic neopentylglycol esters in the presence of $Ru_3(CO)_{12}$ catalyst to give 2-substituted 5-arylpyrrolines in up to 76% yield albeit as a *cis*/*trans* mixture. Arylboronic neopentylglycol esters containing CF_3, Ac, OMe, and Me

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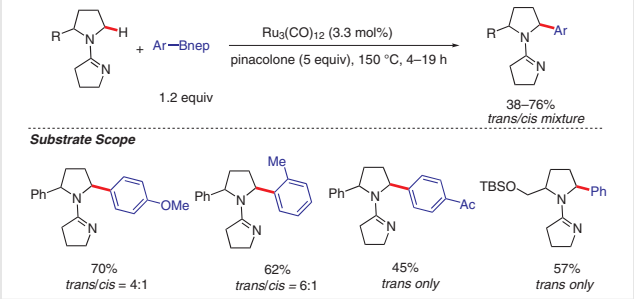
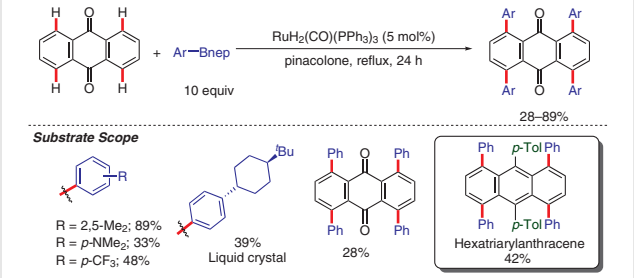
groups gave the corresponding products in good yields, and indolyl- and pyridylboronates reacted with satisfying yields. Generally the products were obtained with 6:1 to 3:1 *trans*/*cis* ratio; a few products were obtained solely as the *trans* isomer. It was shown that if the purified *trans* or *cis* isomer was subjected to the optimized conditions, this resulted in rapid interconversion within 2 hours which was complete in ~24 hours that gave a *cis*/*trans* mixture that was the same as that observed from the original arylation experiments.



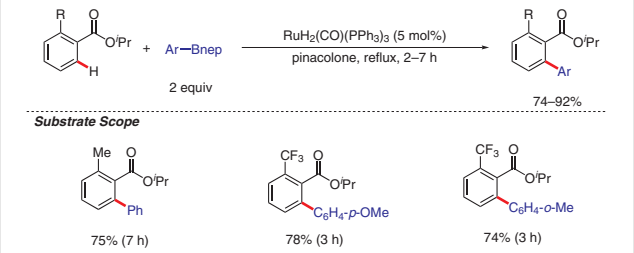
Scheme 40 sp^3-C–H arylation of pyrrolidines and piperidines through Ru catalysis directed by cyclic imine functionality

In 2009, the Kakiuchi group utilized their previously developed $RuH_2(CO)(PPh_3)_3$ catalyst (Scheme 39)[52] to directly arylate anthraquinone and further synthesis of linearly fused PAHs, chelated by carbonyl functionality (Scheme 41).[55] Arylboronic neopentylglycol esters substituted with Me, NMe_2, hexyl, OMe, and CF_3 underwent coupling with ease with anthraquinone and produced moderate to excellent yields of tetrarylanthraquinones in up to 89% yield. 1,4,5,8-Tetra(4-tolyl)anthracene-9,10-dione was reduced in the presence of HI (10 equiv) and aqueous AcOH at 140 °C to form 1,4,5,8-tetra(4-tolyl)anthracene in 83% yield. Spatially crinkled hexaarylanthracenes were also produced in 36–47% yield by a reduction strategy that introduced two aryl groups at C-9 and C-10 using ArLi then NaI/NaH_2PO_2, which could allow the tuning of spectral as well as spatial properties. Interestingly, X-ray crystallographic studies of the molecule revealed its end-to-end twisted puckered structure.



Scheme 41 Polyarylation of multiple C–H bonds of anthraquinone through Ru catalysis assisted by ketone group ligation

In 2010, the Kakiuchi group extended their previous work[52] on aryl ketones (Scheme 39) to the *ortho*-arylation of benzoates, utilizing direction by the ester moiety, with arylboronic neopentylglycol esters using their previously developed Ru catalyst[52] to give to give biaryls (Scheme 42).[54] The presence of electron-withdrawing groups on the benzoate improved the product yield whereas electron-donating groups gave lower yields. This electronic effect is the opposite to that for the arylation of aryl ketones where electron-donating groups produced better yields. Benzoates containing CF_3 and arylboronic neopentylglycol esters containing OMe and Me were all reacted successfully. The biphenyl-2-carboxylic esters were hydrolyzed to give biphenyl-2-carboxylic acids in up to 93% yield thus demonstrating the synthetic utility of the method.



Scheme 42 Ru-catalyzed *ortho*-C–H arylation of arenes directed by ester group chelation

In 2011, Wan and co-workers described the pyridyl and pyrazolyl group directed *ortho*-arylation of 2-(2-substituted phenyl)pyridines and 1-(2-substituted phenyl)pyrazoles with boronic acids using molecular oxygen as the terminal oxidant, $[Ru(p\text{-cymene})Cl_2]_2$ as the catalyst, and $BiBr_3$ as an additive (Scheme 43).[51] Various boronic acids having *t*-Bu, Me, OMe, CF_3, halo, and OCF_3 functionalities were well tolerated giving the arylation products in up to 92% yield. 1-(3-Nitrophenyl)pyrazole containing an electron-deficient arenes failed to react with phenylboronic acid under the optimized conditions. Notably, (2-substituted phenyl)pyridines and 1-(2-substituted phenyl)pyrazoles required *ortho*-substitution by Me or Ph groups for exclusive monoarylation. The additive $BiBr_3$ was proposed to act as a co-oxidant in tandem with oxygen in the final step. Experimental studies suggested a KIE value of 1.5 implying C–H bond cleavage in the rate-determining step.

The proposed mechanism (Scheme 43), based upon experimental evidence, begins with directing group chelated *ortho*-metalation and base-mediated deprotonation to form **C1**. This is followed by transmetalation of **C1** with the arylboronic acid to give **C2**, which undergoes reductive elimination to give the product and Ru(0) that is oxidized cooperatively by $BiBr_3$ and oxygen to regenerate the metal catalyst.

In 2012, Schnürch and co-workers reported the arylation of benzylic amines directed by a 3-substituted 2-pyridyl anchor using arylboronic trimethyleneglycol esters

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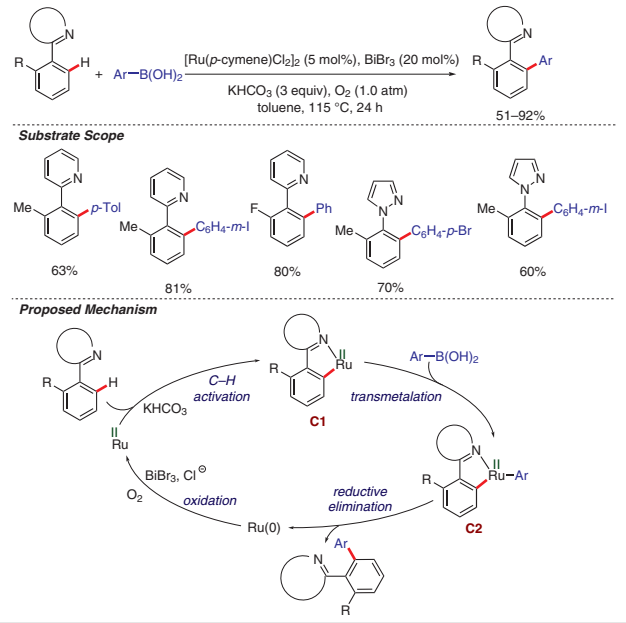
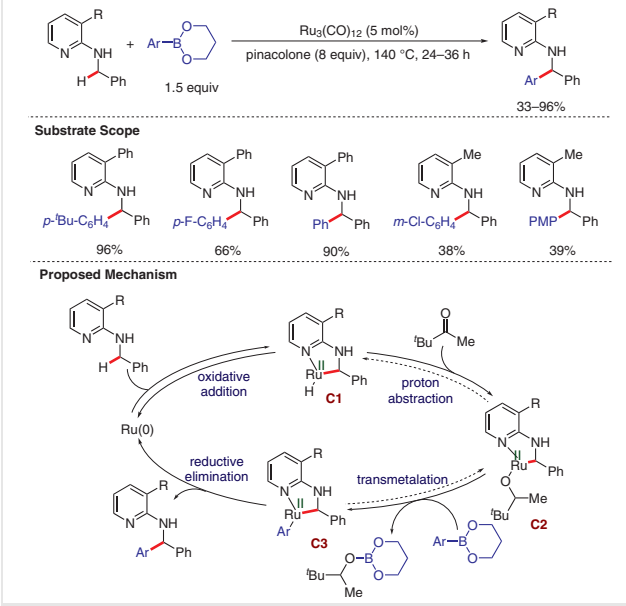


Scheme 43 Ru-catalyzed *ortho*-C–H arylation directed by pyridyl and pyrazolyl group ligation and proposed mechanism



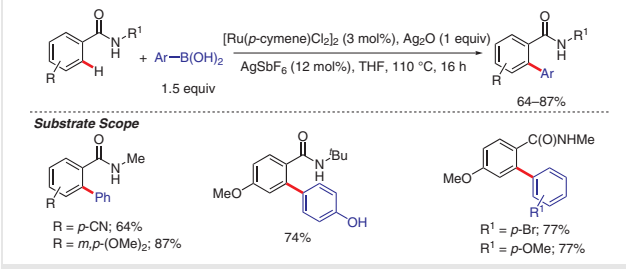
Scheme 44 Ru-catalyzed sp^3-C–H arylation of benzylic amines assisted by substituted pyridyl group direction and proposed mechanism

(Scheme 44).[56] It was essential that the 2-pyridyl group contained a 3-substituent (Me or Ph) to facilitate reaction by ensuring the benzylic sp^3-C–H pointed away from the pyridine nitrogen. The 2-pyridyl directing group was removed in one example by a two-step protocol that involved first protection of the amine by the Boc group and then hydrolysis to give the *N*-Boc-diarylmethanamine in 92% and 91% yields respectively. Arylboronic trimethyleneglycol esters substituted at the 3 position by Me and Cl and the 4 position by Me, F, Cl, CF_3, *t*-Bu, OMe, Ac, CN, and NO_2 moieties were compatible, however reaction of 2-tolylboronic trimethyleneglycol ester was unsuccessful. Mechanistic investigations revealed evidence of 'inverse' KIE, an observation that indicates reversibility of the C–H scission step and the pathway involves a σ-complex.

Based on the work by Kakiuchi and co-workers,[52] the proposed mechanism (Scheme 44) involves initial coordination of Ru(0) with the pyridine nitrogen and deprotonation to form **C1**. Arguably, the oxidative addition step is reversible, which was shown by the inverse KIE. Next, pinacolone acts as a sacrificial hydride scavenger forming alkoxy-ruthenacycle **C2**, which was detected by GS-MS studies. **C2** undergoes transmetalation to give **C3** that undergoes reductive elimination to give the product and regenerating Ru(0).

In 2012, Jeganmohan and Chinnagolla investigated arylation and alkenylation processes directed by *N*-alkylbenzamides (Scheme 45).[59] Thus the arylation of *N*-arylbenzamides with arylboronic acids using [Ru(*p*-cymene)Cl_2]$_2$ as the catalyst with $AgSbF_6$ and AgO in THF at 110 °C gave 2-arylbenzamides in 64–87% yields. The presence of Ag_2O and $AgSbF_6$ as additives was mandatory; Ag_2O is thought to be a co-oxidant and a base in the process, but the role of $AgSbF_6$ remained unidentified. Observably, arylation takes place at a less congested C–H bond on benzamide in the presence of bulky functionalities on the ring to avoid steric interaction between existing and incoming groups. Benzamides with I, Br, NO_2, CN, and Me, and arylboronic acids with OH, OMe, Ac, Br, and F all reacted successfully under the optimized conditions. In two examples, the 2-arylated benzamides were subjected to intramolecular annulation in the presence of TFAA and HCl to give 9-fluorenones in up to 89% yield, thus demonstrating the utility of the method.

In 2014, Jeganmohan and Chinnagolla developed another method for exclusive monoarylation of symmetrical acetanilides, isomeric to benzamides, that seldom generate 2-arylacetanilides (Scheme 46).[60] Such anilides are an integral part of several bioactive molecules like bixafen, boscalid, fluxapyroxad etc.[61] In contrast to their previous study[59] which utilized the assistance of the amide nitrogen, the



Scheme 45 *ortho*-C–H arylation of benzamides by Ru catalysis

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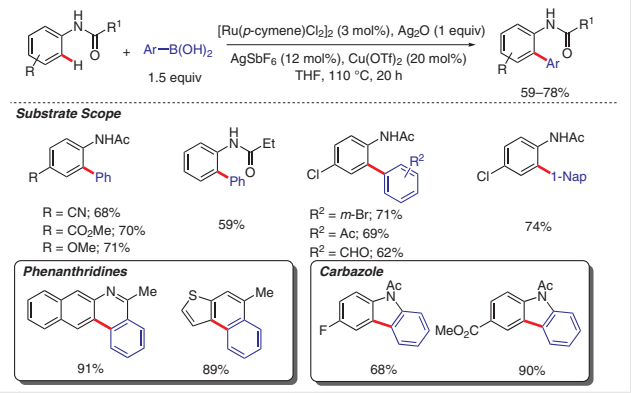
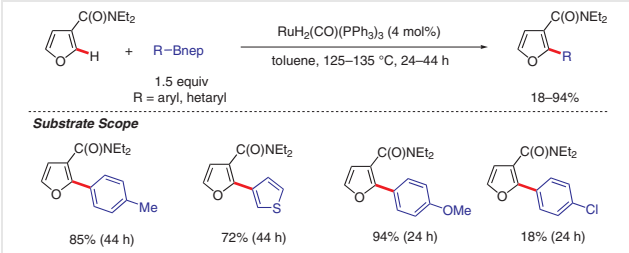
carbonyl on the acetamide was observed to act as the *ortho*-directing anchor in this method in the presence of $[Ru(p\text{-cymene})Cl_2]_2$ catalyst and Ag_2O, $AgSbF_6$, and $Cu(OTf)_2$ as additives In this reaction, $Cu(OTf)_2$ eased C–H scission and stabilized the catalyst in the presence of the Ag^+ ion. Various functionalities were accommodated on both acetanilides and boronic acids under the optimized conditions. A number of carbazoles and phenanthridines were synthesized by the intermolecular annulation of the arylated products in up to 97% yield, thus demonstrating the pharmaceutical importance of the method.



Scheme 46 Ru-catalyzed *ortho*-C–H arylation of substituted arenes directed by amide group ligation and subsequent phenanthridine and carbazole synthesis

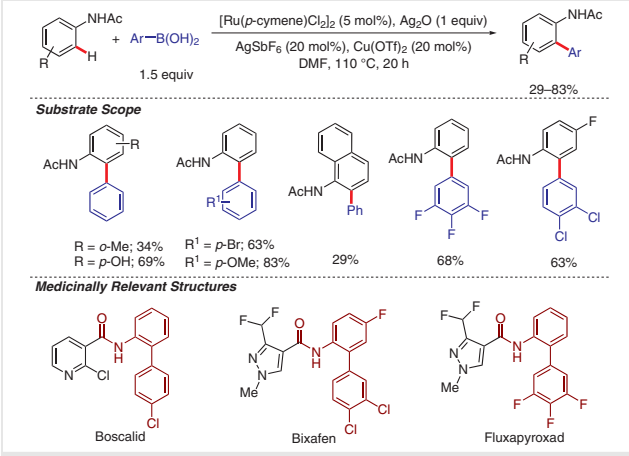
In 2014, Snieckus and Zhao reported the (het)arylation that used a tertiary amide group to regioselectively (het)arylate *N,N*-diethylfuran-3-carboxamide with arylboronic neopentylglycol esters at C-2 using $RuH_2(CO)(PPh_3)_3$ (Scheme 47).[62] The arylboronic neopentylglycol esters were functionalized by a range of functionalities (OMe, Me, F, NMe_3, CF_3, CHO) and under the optimized conditions gave the 3-aryl-*N,N*-diethylfuran-2-carboxamides in up to 94% yield. Only unsubstituted *N,N*-diethylfuran-3-carboxamide and -2-carboxamide were used, but *N,N*-diethylcarboxamide-substituted thiophene, pyrazine, pyridine indole and thiophene were successfully arylated. Also, 2-furyl, 3-thienyl-, and 2-benzofurylboronic neopentylglycol esters, amongst others, were coupled to *N,N*-diethylfuran-3-carboxamide in moderate to good yields.

Following the work of Jeganmohan and Chinnagolla (Scheme 46),[60] the Ackermann group extended the scope of the arylating agent from arylboronic acids to diarylborinic acids and potassium aryltrifluoroborates for the arylation of acetanilides (Scheme 48).[61] Observably, the anilides were only productive when an acidic N–H bond existed in the acetamide anchor as tertiary amides failed to react. The optimized conditions used $[Ru(p\text{-cymene})Cl_2]_2$ as the catalyst with $AgSbF_4$ and $Cu(OTf)_2$ as additives and Ag_2O the oxidant



Scheme 47 Ru-catalyzed C–H arylation of furans assisted by tertiary amide chelation

in DMF at 110 °C, generally good yields were achieved, but there was considerable dependence upon the nature of the solvent used. Arylboronic acids substituted with Ac, OMe, Me, and halo groups and acetanilides with CO_2Me, Ph, halo, and OH groups gave the products in up to 83% yield. Diarylborinic acids and potassium aryltrifluoroborates coupled with comparable efficiency.



Scheme 48 Ru-catalyzed *ortho*-C–H arylation of substituted arenes directed by amide group chelation and relevant pharmacophoric architectures

In 2015, the Maheswaran group undertook the *ortho*-arylation of arenes containing pyridyl and pyrazolyl directing groups (Scheme 49).[63] For a reaction using similar substrates and conditions see Wan and co-workers (Scheme 43).[51] This reaction required the use of 1.2 equivalents of the arylboronic acid; greater loading of the arylating agent led to unwanted diarylation. The optimized conditions used $[Ru(p\text{-cymene})Cl_2]_2$ as the catalyst and $PhI(OCOCF_3)_2$ as the oxidant in toluene in the absence of an external base. Arylboronic acids and 2-arylpyridines or 1-arylpyrazoles with a range of functionalities were successfully utilized in up to 92% yield (2-pyridyl anchor) and up to 78% yield (pyrazol-1-yl anchor). Control experiments revealed the possibility of a SET mechanism playing a central role in this transformation.

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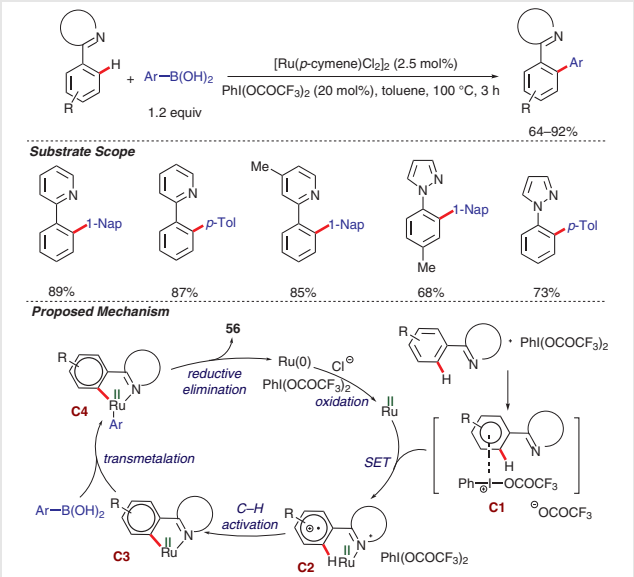
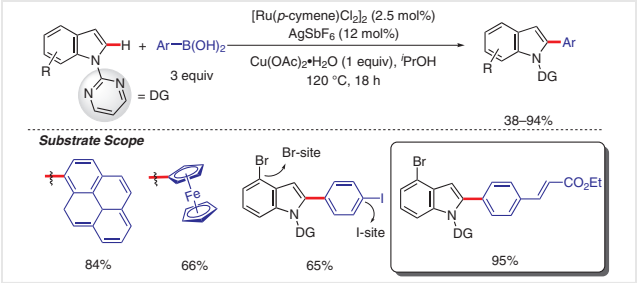


Scheme 49 Substituted arene *ortho*-C–H arylation through Ru catalysis with 2-pyridyl and pyrazol-1-yl group assistance and proposed mechanism

According to the proposed mechanism (Scheme 49), well-known charge-transfer complex **C1** is produced through coordination between $PhI(OCOCF_3)_2$ and the arene of the 2-arylpyridine or 1-arylpyrazole. **C1** reacts with Ru(II) following a SET pathway, forming a radical cation species **C2**. Due to proximal interactions, cyclometalation happens at the *ortho*-position of **C2** through C–H activation to give **C3** that subsequently undergoes transmetalation with the arylboronic acid forming **C4**. Reductive elimination from **C4** gives the product and Ru(0). $PhI(OCOCF_3)_2$ then oxidizes Ru(0) to regenerate the catalyst to further the catalysis cycle.

Also in 2015, Pilarski and co-workers reported the use of a 2-pyrimidyl direction group. The arylation of *N*-(2-pyrimidyl)indoles with arylboronic acids using the $[Ru(p\text{-cymene})Cl_2]_2/AgSbF_6/Cu(OAc)_2{\cdot}H_2O$ system, without external base, in *i*-PrOH gave 2-aryl-*N*-(2-pyrimidyl)indoles in up to 94% yield (Scheme 50).[64] *i*-PrOH was the optimized solvent for this transformation, however in some cases the use of THF/water improved the yield, presumably due to the formation of alkoxyboronates in situ, giving improved nucleophile solubility and slower protodeborylation. Arylboronic acids substituted by NO_2, Ac, SO_2Me, ferrocenyl, and halo functionalities performed well, but the use of protected boronic acids, like ArBpin and $ArBF_3K$, was unsuccessful under the optimized conditions. 3-Methyl-*N*-(2-pyrimidyl)indole underwent the reaction with phenylboronic acid under the optimized conditions to give the corresponding product in 75% yield despite the proximate steric bulk. Demonstrating biological applicability, an *N*-phthaloyl-protected 1-(2-pyrimidyl)tryptophan, a proteinic amino acid, was arylated using ferrocenylboronic acid in 76% yield. (Un)substituted *N*-(2-pyrimidyl)pyrroles were also valuable substrates under the optimized conditions giving 2-arylated *N*-(2-pyrimidyl)pyrroles, in up to 80% yield. The arylated product 4-bromo-2-(4-iodophenyl)-*N*-(2-pyrimidyl)indole containing a Br and I moiety on each ring underwent Heck-type alkenylation only at the iodo site, which suggests late-stage functionalization possibilities.



Scheme 50 C–H arylation of *N*-substituted indoles through Ru catalysis assisted by pyrimidyl group chelation

In 2016, Szostak and co-workers exploited a tertiary amide anchor for the *ortho*-arylation of benzamides and showed that the weakly coordinating carbonyl functionality is the directing group (Scheme 51).[65] Directed *ortho*-metalation requires strong lithium bases and low temperatures that reduce chemodivergence and regioselection. In contrast, this approach avoided the use of an external base and accommodated functionalities, such as CHO, ester, Ac, halo, CF_3, NO_2, etc. Arylated *N*,*N*-dialkylbenzamide units make up the elementary skeleton of fungicides, orexin receptor antagonists, and antiandrogenic agents, implying the pharmacological significance of the method. Reaction of the tertiary amides *N*-(arylcarbonyl)piperidine and -morpholine with arylboronic acids under the optimized conditions using $[Ru(p\text{-cymene})Cl_2]_2/AgSbF_6/Ag_2O/Cu(OTf)_2/H_2O$ in THF gave the corresponding 2-arylated products in up to 90% yield. The reaction was performed on a gram scale giving the arylated product in 89% yield. Mechanistic investigations revealed the C–H scission step to be reversible, and the formation of a σ-complex formation was proposed.

The Kakiuchi group expanded further their original 2005 work (Scheme 39)[52] and in 2017, reported the use of their catalyst system for the arylation of substituted electron-deficient arenes with the cyano group as the directing anchor (Scheme 52).[57] The use of the $RuH_2(CO)[P(4\text{-Tol})_3]$ catalyst provided higher yields of the *ortho*-arylation products than the conventional $RuH_2(CO)(PPh_3)_3$ catalyst. The arylation takes place mostly at the *ortho* positions, but unprecedented *para*-arylation was also partially observed to give *ortho*, *para*-diarylation products. The arylation reaction was unsuccessful when the cyano group was replaced by the trifluoromethyl group, indicating the paramount significance of cyano presence. Arylboronic neopentylglycol esters substituted with OMe, Me, hexyl, NMe_2, F, and CF_3 gave the corresponding products in up to 85% yield, but

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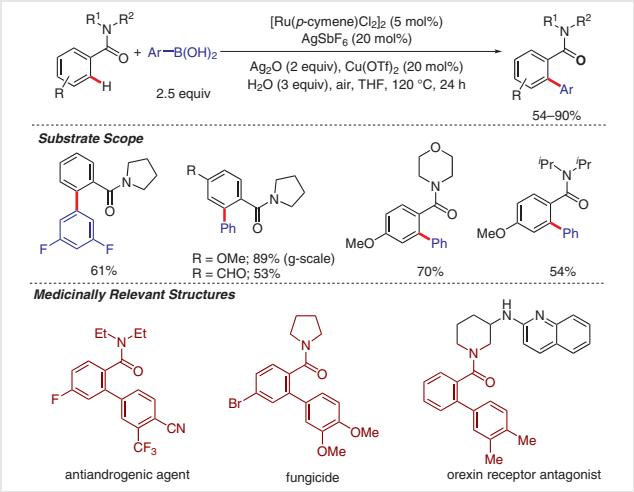
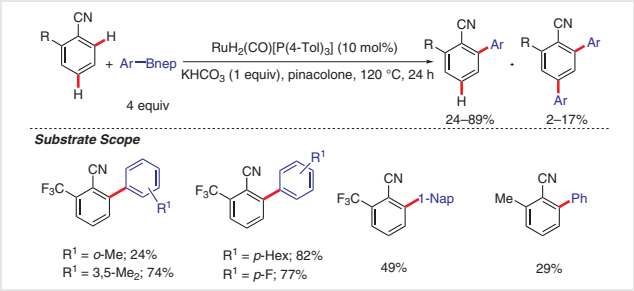


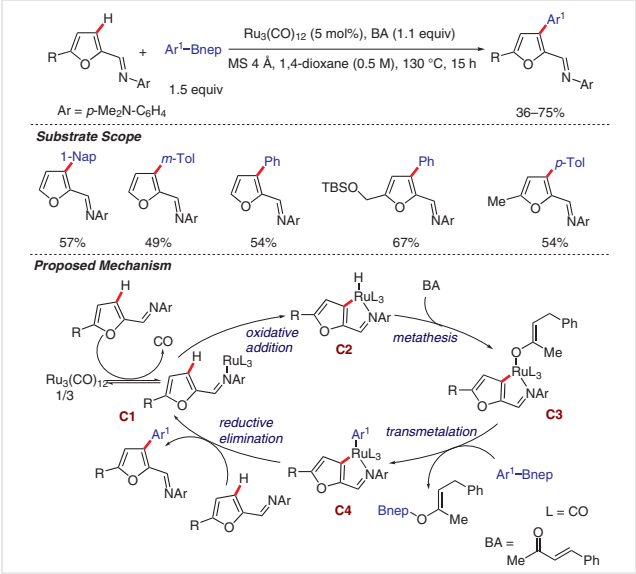
Scheme 51 Ru-catalyzed *ortho*-C–H arylation of substituted arenes assisted by tertiary amide ligation

sterically encumbered 2-tolyl- and 1-naphthylboronates produced inferior yields, thus showing the sensitivity of the method to steric hindrance. Although there was no conclusive mechanistic pathway proposed, controlled experiments reiterated the cleavage of the C–H bond in the rate-determining step.

In 2018, Poli and co-workers reported the arylation of furfural imines, using arylboronic neopentylglycol esters with the imine functionality behaving as a directing group (Scheme 53).[66] (Un)substituted furfural, a feedstock component of unsaturated biomass is easily obtained from cyclodehydration of natural carbohydrates. Interestingly, the in situ formation of Ru–H was observed, and this reduced the directing group, however the application of a sacrificial hydride scavenger,[52] benzylideneacetone (BA), resolved the undesired reduction problem. The use of *N*-aryl-2-furylmethanimines containing electron-donating groups on the *N*-arylimine functionality gave better yields, implying the enhanced nucleophilicity of the nitrogen and subsequent beneficial coordination with the Ru catalyst. A wide range of functional groups on the arylboronic neopentylglycol esters were compatible with the optimized protocol, except for 2-tolylboronic neopentylglycol ester. The diarylation of a bioactive 5,5′-[oxybis(methylene)]bis(furan-2-carbaldehyde) diimine in 47% yield illustrates the utility of the method.



Scheme 52 *ortho*-C–H arylation of substituted arenes through Ru catalysis aided by CN group direction



Scheme 53 C-3 C–H arylation of furfurals by Ru catalysis assisted by imine group ligation and proposed mechanism

A mechanism was proposed based on the literature (Scheme 53). Initially, the Ru–Ru bond in $Ru_3(CO)_{12}$ is cleaved and Ru is chelated by the imine nitrogen forming **C1**. Then, cyclometalation results in ruthenation at C-3 of the furyl group and Ru–H is formed in **C2**. The hydride is abstracted by benzylideneacetone and the Ru–H is replaced by Ru–alkoxy producing **C3**. Upon transmetalation, reduced benzylideneacetone also behaves as the boron scavenger and aided in transmetalation of **C3** to generate **C4**. Finally, **C4** undergoes reductive elimination and subsequent transimination in the presence of another furyl imine to regenerate **C1** and give the final product.

In 2019, Chang and co-workers reported a method for the arylation of quinoline *N*-oxides where the *N*-oxide behaved as a directing group for the arylation at C-8 using arylboronic ethyleneglycol esters and subsequent deoxygenation of the *N*-oxide by the Ru catalyst to produce 8-arylquinolines (Scheme 54).[67] Using the optimized conditions of $[Ru(p\text{-cymene})Cl_2]_2/AgOTs/Ag_2O/CuClO_4{\cdot}6H_2O$ in THF/TFE solvent, it was proposed that 2,2,2-trifluoroethanol (TFE) also assists in the Ru-catalyzed *N*-oxide reduction. Using Ag_2O as an oxidant and $Cu(ClO_4)_2{\cdot}6H_2O$ as transmetalation promoter did not produce the deoxygenated product in the absence of the Ru catalyst. Quinoline *N*-oxides containing synthetically manipulative functionalities, like CO_2Me, Ac, CHO, Ac, OSi(*i*-Pr)$_3$, and NHAc gave 8-arylquino-

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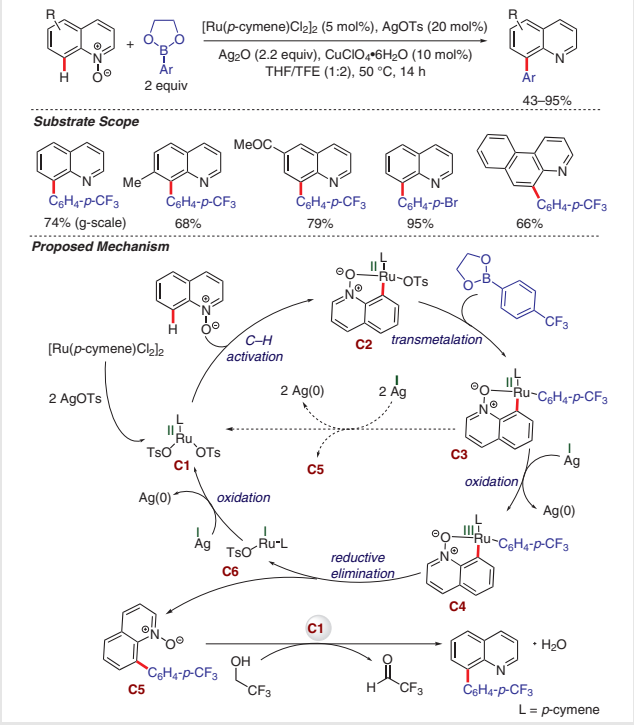
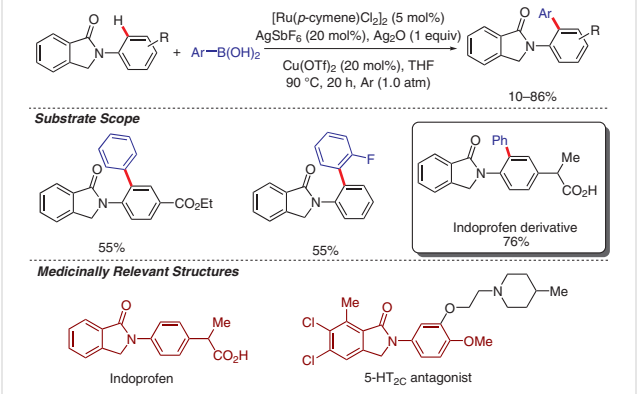
lines in up to 93% yield. Mechanistic investigations revealed negligible KIE, advocating the absence of C–H cleavage in the rate-determining step.



Scheme 54 C–H arylation of quinoline *N*-oxides through Ru catalysis with *N*-oxide group ligation assistance and proposed mechanism

The proposed mechanism (Scheme 54) illustrates arylation and simultaneous deoxygenation. Initially, the precatalyst produces ruthenium species **C1**, the active catalyst, which coordinates to the O-atom of the *N*-oxide followed by deprotonation through cyclometalation to form **C2**. Then, transmetalation of **C2** results in the aryl–Ru complex **C3**, and Ag(I) oxidizes the metal center in **C3** to give **C4** which undergoes reductive elimination to produce **C5**. In presence of ruthenium species **C1**, **C5** undergoes deoxygenation and the directing functionality is eliminated by oxidizing TFE. Intriguingly, it is thought that direct elimination might take place, liberating the intermediate **C5** upon oxidation by Ag(I) albeit no evidence was presented to support this.

Gramage-Doria and co-workers utilized conditions similar to those used by Szostak and co-workers (Scheme 51)[65] for the arylation of *N*-arylisoindolin-1-ones, containing a weakly coordinating carbonyl group, with arylboronic acids to give *N*-biphenyl-2-ylisoindolin-1-ones (Scheme 55).[68] In contrast to the work of Szostak and co-workers, this method *ortho*-arylated the aryl group on the amide nitrogen and did not require the use of water. Arylboronic acids were better arylating agent that to their protected ester counterparts and reactions with OMe, NO_2, and halo group substituted arylboronic acids were successful with the exception of sterically hindered 2-tolylboronic acid. Interestingly, the presence of a carboxylate group on the aryl group on the amide nitrogen did not exert a directing influence, implying the precise regioselective nature of the method. Notably, reactions of heteroarylboronic acids were unsuccessful and the reaction was also unsuccessful using *N*-arylisoindoline which lacked a carbonyl group. Illustrating late-stage functionalization possibilities, indoprofen was derivatized through esterification, direct arylation, and ester hydrolysis in 76% overall yield (3 steps). Although masking of the carboxyl group of indoprofen appeared necessary due to incompatibility toward the optimized conditions, the exceptional derivatization yield spoke for the method developed.



Scheme 55 *ortho*-C–H arylation of substituted arenes through Ru catalysis assisted by ketone group chelation and relevant pharmacophoric architectures

In 2020, Sharma and co-workers reported the arylation of 8-methylquinolines (8-QM) using arylboronic acids exploiting $[Ru(p\text{-cymene})Cl_2]_2$ as the catalyst albeit in moderate yields in most cases (Scheme 56).[69] The arylboronic acid could be substituted by Br, NO_2, CF_3, *t*-Bu, OMe, and $SiMe_3$ groups while the 8-methylquinoline could be substituted by halogen to give the 8-benzylquinoline products in up to 52% yield. Speculatively, methyl sp^3-C–H arylation receives some assistance from the quinoline nitrogen as the proposed mechanism documented a ligation with the metal. Mechanistic investigations revealed the irreversibility of C–H scission and its absence in the rate-determining step. The reaction was performed on a gram scale, 8-methylquinoline was arylated with 4-(trimethylsilyl)phenylboronic acid in 32% yield. Notably, scanning electron microscopy (SEM) studies did not detect any Ru nanoparticles, implying a complete homogenously catalyzed transformation.

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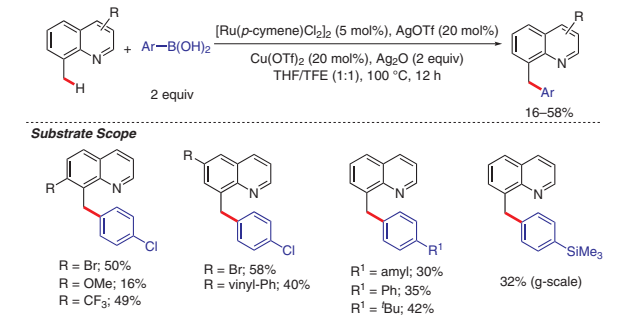
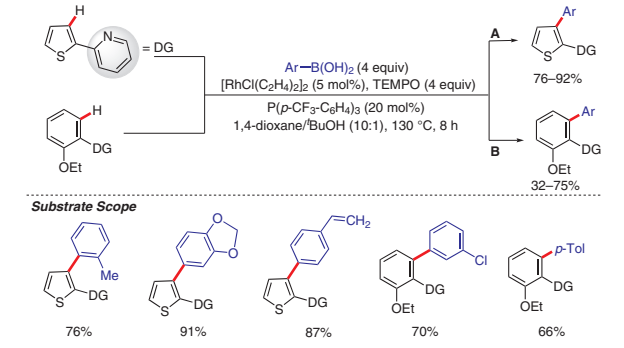


Scheme 56 Ru-catalyzed C–H arylation of substituted 8-methylquinolines at C-8 aided by internal N-atom ligation

3.3 Rhodium Catalysis

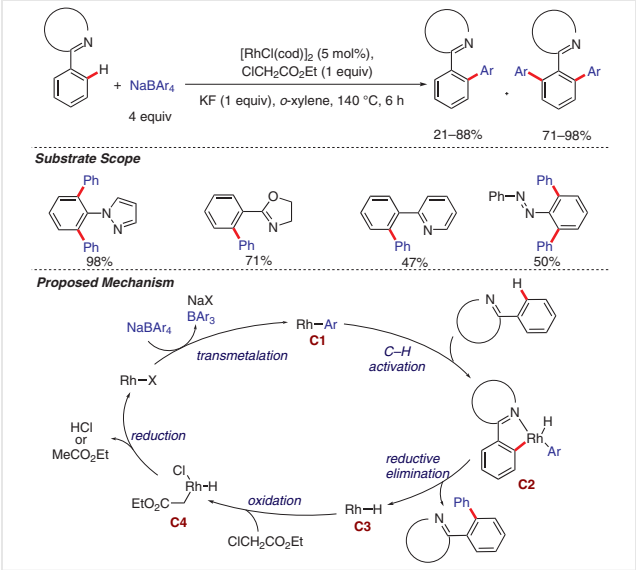
Prior to pursuing the inert C-4 arylation of thiophenes (Section 2.1, Scheme 8),[22] Studer and Vogler investigated the $[RhCl(C_2H_4)_2]_2$-catalyzed pyridyl group directed C-2 arylation of thiophene and *ortho*-arylation of arenes albeit with limited substrate scope (Scheme 57).[70] Interestingly, the electronic nature of the arylboronic acid exerted little impact on the yield; the arylation of thiophenes occurred in up to 92% yield while the arylation of arenes occurred in up to 75% yield. In this reaction, TEMPO behaves as an oxidant as well as a base, and its replacement with oxygen provided inferior yields. In an attempt to replace the pyridyl directing group with a synthetically manipulative directing group, imines were subjected to the optimized conditions and arylated imine products were hydrolyzed to the corresponding aldehydes in up to 51% yield.

In 2008, Miura, Satoh, and co-workers reported the synthesis of various substituted phenylazoles, where the azole behaves as the chelating functionality, with sodium tetraarylborates using $[RhCl(cod)]_2$ as the catalyst (Scheme 58).[71] Initial experiments using benzophenone imines gave a reaction mixture containing a considerable amount of amine and it was speculated that Rh–H formation led to the reduction of the directing imine functionality. An effective solution to this problem was the use of ethyl chloroacetate as a



Scheme 57 Rh-catalyzed *ortho*- and C-2 C–H arylation of thiophenes and arenes assisted by pyridyl group chelation

sacrificial hydride scavenger. The use of KF appeared to facilitate the catalysis, but its primary role in improving the yield remained unidentified. A few azole directing groups directed the mono- and diarylation in good yields for a limited substrate scope.



Scheme 58 *ortho*-C–H arylation of unsubstituted arenes using Rh catalysis with azolyl group assistance and proposed mechanism

Based on the literature, a mechanistic pathway was proposed (Scheme 59) involving initial transmetalation to form **C1** and that intermediate reacts with the azolylarene to produce **C2** upon activating the *ortho*-C–H bond. Reductive elimination on **C2** liberates the product, leaving Rh–H **C3** that is oxidized by $ClCH_2CO_2Et$ to regenerate the active catalyst with elimination of HCl or ethyl acetate. A uniform oxidation state for Rh was proposed but in every step, Rh(I) presence appears to be less likely due to recurring coordination that led the catalysis forward.

In 2012, Cheng and co-workers described the one-pot synthesis of phenanthridones by $[RhCp^*Cl_2]_2$-catalyzed direct arylation of *N*-methoxybenzamides with arylboronic acids and subsequent intramolecular annulation through C–H amination (Scheme 59);[72] this reaction was the first example of dual C–H activation in this context. *N*-Methoxybenzamides substituted with *t*-Bu, Ac, halo, Ph, CF_3, CO_2Me, and NO_2, and arylboronic acids with OMe, Ph, and Me functionalities reacted successfully under the optimized conditions ($[RhCp^*Cl_2]_2$, Ag_2O, MeOH, 60 °C) in up to 94% yield. Mechanistic investigations suggest that it is improbable that C–H activation occurs in the rate-determining step and this led to speculation on prior phenyl ring coordination with the Rh(III) center.

The proposed mechanism (Scheme 59) accounts for the arylation and intermolecular annulation. Removal of Cl^- from $[RhCp^*Cl_2]_2$ by Ag(I) generates the active Rh(III) cata-

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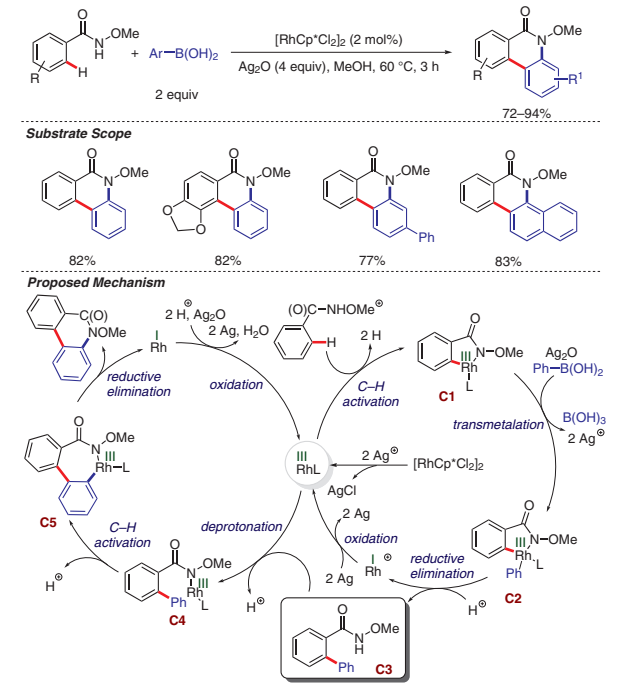
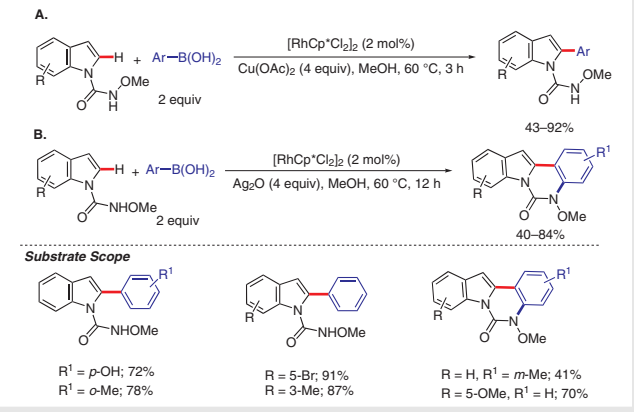


Scheme 59 Phenanthridones by tandem C–H arylation-amination strategy through Rh catalysis with amide group chelation and proposed mechanism

lyst which is chelated by the directing group and subsequently undergoes *ortho*-metalation to generate **C1**. Rhodacycle **C1** undergoes transmetalation by the arylboronic acid to form **C2** which undergoes reductive elimination generating **C3** and leaving Rh(I) that is oxidized by Ag(I) to regenerate the Rh(III) catalyst. Once again, the directing group in **C3** coordinates with the Rh(III) species by liberating a proton, and **C4** paves the way for the formation of rhodacycle **C5** through *ortho*-C–H activation of the installed aryl ring. Subsequently, reductive elimination occurs that to give the phenanthridone and Rh(I) which is reoxidized by Ag(I) to regenerate the active catalyst.

In 2014, Cui and co-workers reported the amide-directed arylation of *N*-methoxy-1*H*-indole-1-carboxamides with arylboronic acids with $[RhCp^*Cl_2]_2$ as the catalyst, $Cu(OAc)_2$ as the oxidant, in MeOH at 60 °C to give 2-aryl-*N*-methoxy-1*H*-indole-1-carboxamides which broadened substrate scope from the work of Cheng and co-workers[72] with an improved set of conditions using $Cu(OAc)_2$ as the oxidant instead of Ag_2O (Scheme 60).[73] *N*-Methoxy-1*H*-indole-1-carboxamides substituted with OBn, OMe, and Br, and arylboronic acids substituted with OMe, OH, Cl, Br, and CN reacted successfully under the optimized conditions giving the corresponding products in 43–92% yields. Reacting the *N*-methoxy-1*H*-indole-1-carboxamides under the optimized conditions of Cheng and co-workers[72] using Ag_2O as the oxidant for a prolonged reaction time gave [4+2] cyclization-type products in up to 84% yield. Performing the reaction of *N*-methoxy-1*H*-indole-1-carboxamide with 4-hydroxyphenylboronic acid under the conditions of Cheng and co-workers[72] with Ag_2O as the oxidant in MeOH in a sealed tube at 90 °C gave a [4+1] cyclization-type product in 32% yield; the spiro structure was subjected to an XRD study. Mechanistic investigations suggest that the rate-determining step does not involve C–H fission as 1.34 KIE was observed.



Scheme 60 Rh-catalyzed C–H arylation and intramolecular annulation through amide group chelation assistance

In 2016, a ligand-accelerated *ortho*-arylation strategy was reported by Lu, Sun, and co-workers using *N*-(pentafluorophenyl)benzamides containing the *N*-pentafluorophenylamide directing group and with arylboronic pinacol esters as the arylating agent (Scheme 61).[74] Optimization of the conditions used $[RhCp^*Cl_2]_2$ as the catalyst with Ag_2CO_3 as the oxidant, BINAP as a ligand, K_2CO_3 as base in MeCN at 80 °C for 3 hours under N_2 atmosphere. The optimization of the ligand showed a similar efficiency of at least two mono-protected amino acids[40,44] and 2,2′-bis(diphenylphosphino)-1,1′-binaphthalene (BINAP), but in terms of commercial viability and cost-efficiency, readily available BINAP was the ligand of choice. *N*-(Pentafluorophenyl)benzamides substituted with OMe, *i*-Pr, Ac, CF_3, and NO_2, and arylboronic pinacol esters substituted with CN, F, Me, etc. were successfully utilized and gave the arylated product in up to 81% yield. In one example the 2-aryl-*N*-(pentafluorophenyl)benzamide was converted into its methyl ester in three steps in 83% yield. The *ortho*-arylation was performed on a gram scale with 78% yield.

In 2017, Lu, Sun, and co-workers reported a method that used the weak coordination of the ketone group for *ortho*-arylation (Scheme 62).[75] Thus aryl ketones were arylated with arylboronic pinacol esters with $[RhCp^*(MeCN)_3](SbF_6)_2$ as the catalyst and Ag_2O as the oxidant in DCE at 120 °C to give biphenyl-2-yl ketones. Aryl ketones substituted by electron-donating groups, like Et, *i*-Pr, OMe, Me on the aryl group and with *t*-Bu, Cy, Ph on ketone were accommodated under the optimized conditions. Observably, α-tetralone

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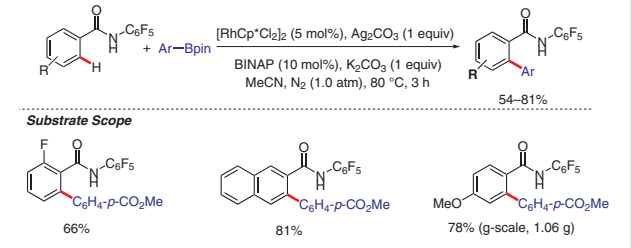
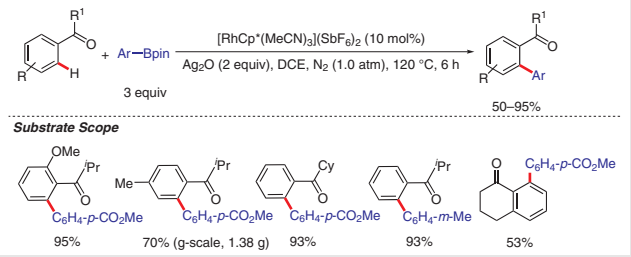


Scheme 61 Rh-catalyzed *ortho*-C–H arylation of substituted arenes directed by amide group ligation

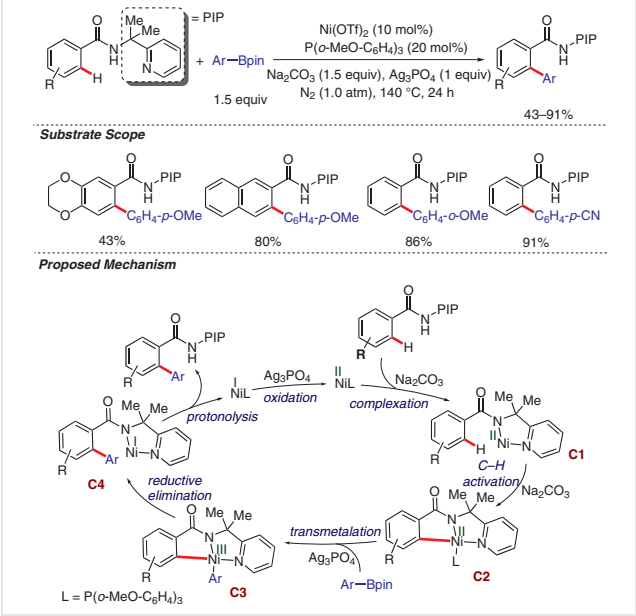
gave 8-[4-(methoxycarbonyl)phenyl]-α-tetralone in only 53% yield, which was attributed to its inflexible and rigid skeleton. Performing the reaction on a large scale gave an astounding 70% yield thus illustrating the scalability of the reaction. Mechanistic investigations revealed an intermolecular KIE of 3.7 and intramolecular KIE of 2.3, implying C–H fission at the rate-determining step.



Scheme 62 *ortho*-C–H arylation of aromatic ketones through Rh catalysis aided by ketone group chelation

3.4 Nickel Catalysis

Instead of using a monodentate directing group, in 2016 B.-F. Shi and co-workers used a bidentate directing group in order to ensure exclusive monoarylation using $Ni(OTf)_2$ as the catalyst (Scheme 63).[76] Using an *N*-2-pyridylisopropyl-substituted (PIP) benzamide turned the monodentate benzamide functionality into a bidentate auxiliary. The optimized conditions for the *ortho*-arylation strategy used $Ni(OTf)_2$ catalyst, with tris(2-methoxyphenyl)phosphine as the ligand, with Na_2CO_3 (base) and Ag_3PO_4 (oxidant) in toluene under a N_2 atmosphere and at a high 140 °C temperature and gave the arylated products in up to 91% yield. The benzamides and arylboronates accommodated a range of synthetically relevant functionalities, including 2-$MeOC_6H_4Bpin$ which gave the corresponding arylated product in 86% yield. The addition of TEMPO to the reaction medium completely inhibited the reaction, prompting the hypothesis of a probable SET mechanism. Further mechanistic investigations suggested C–H scission involvement in the rate-determining step.



Scheme 63 Ni-catalyzed *ortho*-C–H arylation of substituted arenes directed by amide-PIP group ligation and proposed mechanism

The proposed mechanism (Scheme 63) involves Ni(I/II/III) species. Initially, PIP chelates the Ni(II) species to form **C1** followed by *ortho*-C–H activation of the arene giving **C2** which is transmetalated with ArBpin to produce **C3**; and subsequent reductive elimination gives **C4**. Protonolysis of **C4** gives the desired product and a Ni(I) species that is oxidized by Ag_3PO_4 to produce the active catalyst.

3.5 Cobalt Catalysis

The indoline structure is the backbone of many bioactive molecules, for instance dictyodendrin, diazoamide A, dippadine, etc. In 2018, Punniyamurthy and co-workers described the arylation of *N*-(2-pyrimidyl)indolines at C-7 with arylboronic acids, where pyrimidyl functions as a directing group for the ligated metal to give the desired regioselectivity in the presence of $Co(acac)_2$ catalyst (Scheme 64).[77] The use of various functionalities was successful under the optimized conditions although 2-tolylboronic acid and 6-bromo-*N*-(2-pyridyl)indoline did not react for steric reasons. Application of TEMPO reduced the yield to 18%, implying a radical mechanism may be at play. 8-Phenyl-*N*-(2-pyrimidyl)indoline was oxidized to the indole using DDQ and the auxiliary was removed to produce 7-phenylindole in 64% yield over 2 steps.

According to the postulated mechanism (Scheme 64), Co(II) precatalyst is oxidized by Mn(III) which, in turn, is reoxidized by molecular oxygen. **C1** reacts with the *N*-(2-pyrimidyl)indoline through a concerted metalation-deprotonation (CMD) pathway to generate **C2**. Meanwhile, Mn(II) ox-

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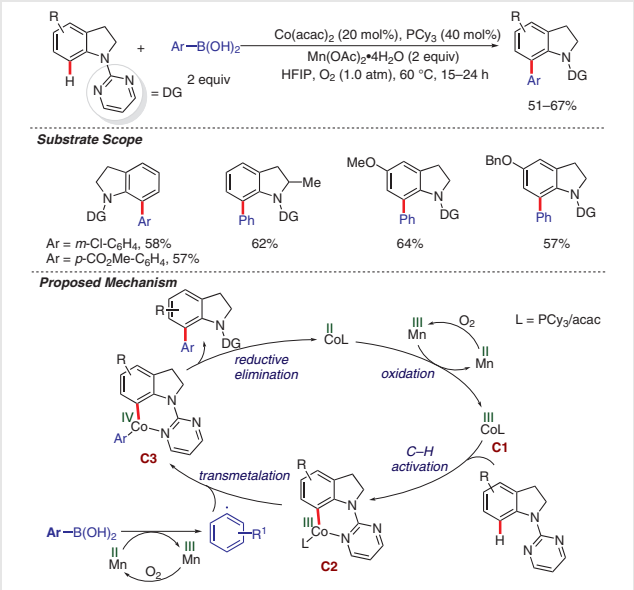
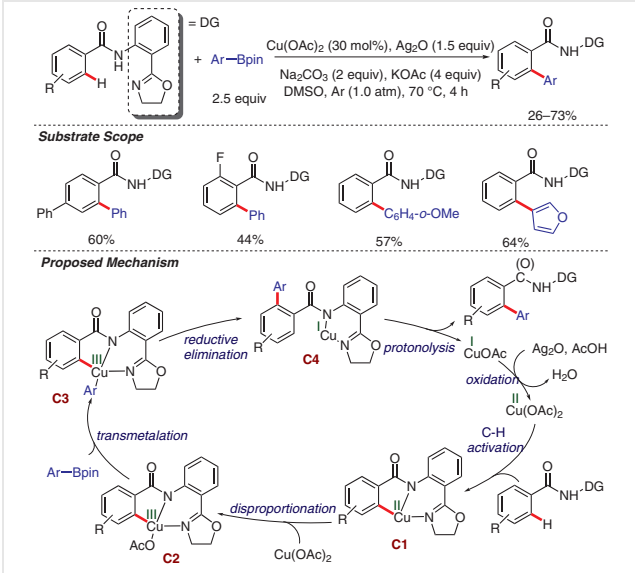


Scheme 64 Co-catalyzed C–H arylation of indolines assisted by pyrimidyl group direction and proposed mechanism



Scheme 65 Cu-catalyzed *ortho*-C–H arylation of substituted arenes aided by amide-oxazoline functionality ligation and proposed mechanism

idizes $ArB(OH)_2$ to form an aryl radical, and this transmetalates **C2** to produce **C3** containing the Co(IV) species. Finally, the product is obtained by reductive elimination and the precatalyst regenerates. A formal reoxidation by Mn(III) fuels the catalytic cycle with the active catalyst.

3.6 Copper Catalysis

In 2014, the Yu group made use of inexpensive $Cu(OAc)_2$ in a catalytic amount to develop an *ortho*-arylation protocol for benzamides using arylboronic pinacol esters (Scheme 65).[78] The amide group of the benzamides required an oxazolyl group to efficiently direct the reaction through bidentate coordination. The presence of electron-donating groups such as OMe, *t*-Bu, Me on the benzamide improved the yield whereas electron-withdrawing groups like Cl, and CF_3 enabled only moderate yields. Notably, heteroarylboronic pinacol esters, containing furyl, pyridyl, etc. were compatible with the conditions albeit generating the products in moderate yields. To demonstrate the synthetic utility of the developed protocol, *N*-[2-(2-oxazolin-2-yl)phenyl]biphenyl-2-carboxamide was protected as the Boc amide (95% yield) then converted into the thiocarboxylate (70% yield), then this was transformed into biphenyl-2-carbaldehyde and -carboxylic acid in 67% and 68% yields, respectively.

A catalytic cycle was proposed (Scheme 65) based on precedent literature. Initially, CuOAc is oxidized in the presence of Ag_2O and the active catalyst reacts with the arene. After chelation, deprotonation leads to C–H activation and *ortho*-metalation occurs producing **C1**. The intermediate, upon disproportionation with $Cu(OAc)_2$ salt possibly forms **C2**, albeit doubts remain about the presence of Cu(III) in the cycle. Transmetalation of **C2** gives **C3** which undergoes reductive elimination to yield the product. CuOAc was regenerated thus completing the catalytic cycle.

4 Conclusion

The ubiquity of aryl rings is a compelling reason for developing synthetic tools to serve the purpose of arylation. To accomplish the task, many metal-catalyzed and metal-mediated methods[79] using organoborons have been strategized amongst which metal-catalyzed reactions have attracted more attention from the synthetic research community and industry. Despite having many merits, metal-catalyzed C–H arylation seldom falls short in providing regioselective outcomes without directing group assistance. Also, the territory of earth-abundant 3d metals in catalytic arylation using arylborons is underexplored and these metals have failed to garner adequate attention. As chemists worldwide remain perpetually thoughtful about the cost-optimization and ecofriendliness of synthetic methods, inexpensive and less toxic 3rd-row transition metals are highly desirable for such straightforward transitions. Nonetheless, organoborons should also play a pivotal role in replacing traditionally used, delicate haloarenes to ease execution and turn methods 'green'. In this regard, this review will be of immense help to understand the behavior of the reactions and identify the existing problems to devise and develop newer strategies with the help of computational studies.

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Conflict of Interest

The authors declare no conflict of interest.

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